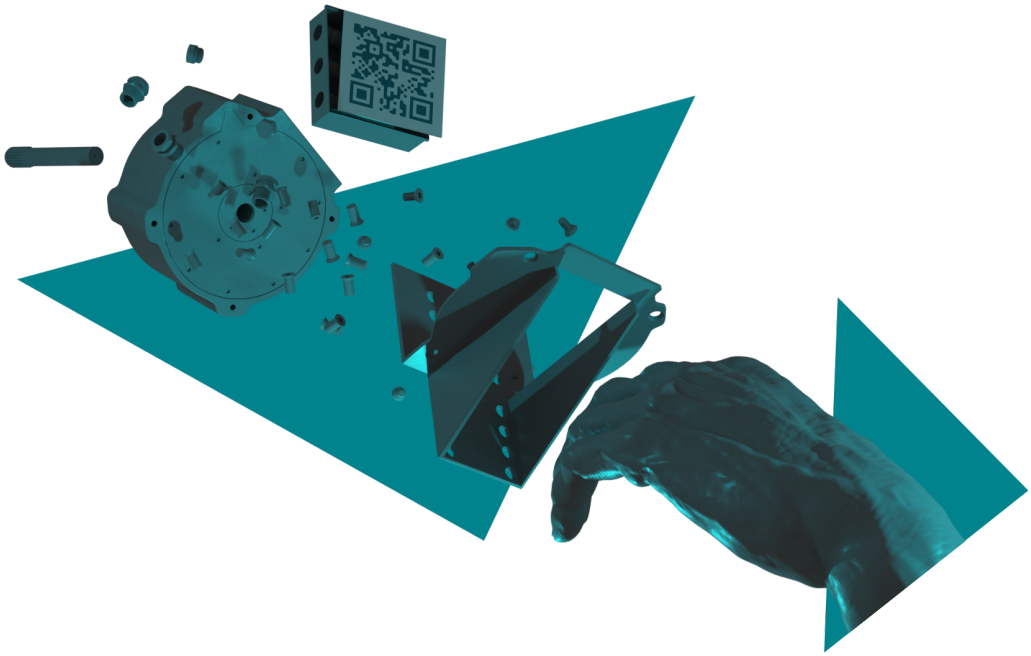
DEGREE PROJECT

Aiding Future Remanufacturing

User-Centered Design of a Digital Product Passport Platform for Remanufacturing of Electric Motors



Mohammad Raad Ghamlouch

Industrial Design Engineering, master's level
2026

Luleå University of Technology
Department of Social Sciences, Technology and Arts

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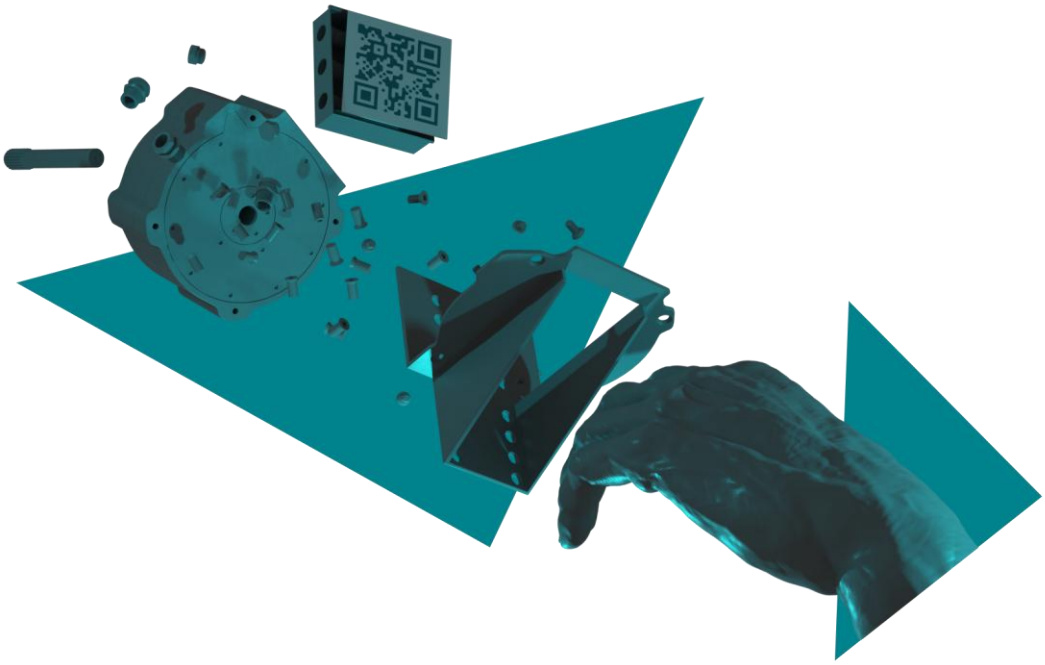
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Aiding Future Remanufacturing

- User-Centered Design of a Digital Product Passport Platform for Remanufacturing of Electric Motors

Mohammad **Raad Ghamlouch**
2025

SUPERVISOR: David **Larsson**
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MSc in INDUSTRIAL DESIGN ENGINEERING
Department of Business Administration, Technology and Social Sciences
Luleå University of Technology

CIVILINGENJÖR I TEKNISK DESIGN
Master of Science Thesis in Industrial Design Engineering

Aiding Future Remanufacturing
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Gothenburg 5th of January 2026
Mohammad Raad Ghamlouch



Abstract

The transition towards a circular economy puts increasing demands on traceability and transparent data accessibility across product lifecycles. Within the European Union, the Digital Product Passport (DPP) has been proposed as a regulatory instrument to allow for sharing of product-related information in support of sustainability. While technical and regulatory aspects of the DPP are currently being developed, less attention has been given to how systems should be designed to support everyday work practices within industrial contexts. The objective of this thesis is to explore how a DPP platform can be designed to support remanufacturing workflows with a user-cantered approach, aiming to bridge theoretical perspectives on usability and change management through practical design exploration. The thesis is conducted within the European Union's Horizon 2020 project RemaNet and focuses on remanufacturing of industrial components, using electric motors as an illustrative use case.

Three alternative design concepts were developed to address different stakeholder perspectives: operational *efficiency* for remanufacturing operators, *(environmental) impact visualization* for managerial stakeholders, and *transparency (communication)* for end users. The concept focused on shop-floor efficiency for DPP logging was developed further into an interactive, mobile-first proof-of-concept prototype. Usability evaluation of the prototype was conducted through cognitive walkthrough-based user testing with six participants from repair and engineering backgrounds. The results indicate that clear workflow alignment, progressive disclosure, task-specific information presentation, and immediate system feedback are vital factors for perceived usefulness and adoption.

The thesis does not propose a fully implemented technical system but contributes to design knowledge, concepts, user feedback, and UI guidelines for how DPP-enabled platforms could be structured to support remanufacturing operations. The intention of the thesis results is to inform future development within the RemaNet project and similar initiatives within the EU.

Keywords: Digital Product Passport, Remanufacturing, User-Centered Design, Usability, Circular Economy, Interaction Design, Industrial Design Engineering

Sammanfattning

Europas övergång till en cirkulär ekonomi betyder ökade krav på spårbarhet, transparens och tillgång till korrekt produkt-data genom hela produktens livscykel. Inom Europeiska unionen har det Digitala Produktpasset (Digital Product Passport, DPP) föreslagits som ett regulatoriskt verktyg för att möjliggöra delning av produktrelaterad information med syftet att stödja hållbarhet. Den pågående utvecklingen av DPP har till stor del fokuserat på tekniska och juridiska aspekter, medans utformningen av hur sådana system bör fungera i vardagliga industriella arbetsprocesser fått mindre uppmärksamhet.

Syftet med detta examensarbete är att undersöka hur en DPP-plattform kan utformas för att stödja återtillverkningsprocesser genom ett användarcentrerat angreppssätt. Målet av arbetet är att genom teoretiska perspektiv på användbarhet, interaktionsdesign och förändringsledning, utforska hur man kan göra en sådan plattform attraktiv för målgruppen. Studien har genomförts inom EU:s Horizon 2020-projekt RemaNet och fokuserar på återtillverkning av industriella produkter, i detta fall elmotorer.

Arbetet följer ett designorienterat ramverk strukturerad kring, s.k. Double Diamond-processen. Metoder som använts inkluderar litteraturstudie, semistrukturerad intervju med en målgrupps-representant, kartläggning av arbetsflöden och kundresor, iterativ idéutveckling, och prototyputveckling via pappersbaserade wireframes och Figma. Tre alternativa designkoncept togs fram för att utforska olika intressentperspektiv: *effektivitet* för återtillverkningsoperatörer, *visualisering av miljöpåverkan* för ledande och hållbarhetsansvariga roller samt *transparens (kommunikation)* för slutkund. Konceptet med fokus på operativ effektivitet vidareutvecklades till en interaktiv, mobil-anpassad proof-of-concept prototyp.

Prototypen utvärderades genom användartester baserade på en kognitiv genomgång (cognitive walkthrough) med sex deltagare verksamma inom reparation- och ingenjörsindustrin. Resultaten visar att tydlig koppling till befintliga arbetsflöden, successiv informationspresentation, uppgiftsspecifika data och omedelbar återkoppling (från systemet) är avgörande faktorer för att användarna ska uppleva nytta, tillit och acceptans. Samtidigt identifierades ett starkt behov av ännu tydligare återkoppling kring progression och systemstatus.

Examensarbetets mål är inte i fullt implementerat tekniskt system, utan att bidra med designkunskap, koncept, användarinsikter och riktlinjer för hur DPP-baserade plattformar kan utformas för att stödja återtillverkning i praktiken. Resultaten är avsedda att fungera som underlag för fortsatt utveckling inom RemaNet-projektet och liknande initiativ inom EU.

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1 Introduction

This master's thesis in Industrial Design Engineering at *Luleå University of Technology* (LTU) is carried out within the European research project *RemaNet*, part of the *Horizon Research and Innovation* program, that seeks to create a federated digital platform for remanufacturing. The platform is based on the *Digital Product Passport* (DPP), which is an integral component of the European Union's strategy for advancing transparent and sustainable, circular economy practices (European Commission, 2025).

This traceable standardized data will be accessible through QR codes or RFID tags, with which EU aims to transform how products are designed, used, maintained and remanufactured (European Commission, 2025). Despite the ongoing development of the DPP's technical infrastructure, the challenge of making the system usable and adoptable persists. Digital tools often fail not because of the lack of need or flawed technology, but because design principles have been overlooked in the human-machine interaction (Norman, 2013). If a system is cumbersome, unclear, or perceived as adding unnecessary work, the users will resist it. For the DPP system to fulfil its potential, it must be not only technically sound, but also intuitive and attractive by those who depend on it.

This thesis will contribute to RemaNet's *Work Package 3* (WP3), which addresses the creation of safe and attractive work environments in remanufacturing (RemaNet, 2024). Focus will be on change management, to promote adoption and minimizing barriers, as well as increasing usability of the platform, built on the DPP system. The project will involve identifying the various stakeholders who interact with the DPP in a manufacturing context, and developing guidelines that ensure the system becomes an attractive facilitator of work, rather than an additional obstacle.

1.1 BACKGROUND

Remanufacturing is a central practice within the circular economy, offering significant environmental and economic benefits by restoring used products to a state comparable to new manufacturing. Despite its potential, remanufacturing is frequently hindered by limited access to reliable product data, such as usage history, disassembly instructions, BOMs, material composition, and environmental impact metrics. These information gaps increase operational difficulty and reduce confidence in remanufacturing outcomes. At the same time, sustainability regulations and reporting requirements are placing increased pressure on companies to quantify and document their environmental impacts, particularly within the European Union. This in turn has created a growing demand for standardized and verifiable lifecycle data that can support both operational decision-making and regulatory compliance.

The Digital Product Passport (DPP) has been proposed as a solution to these challenges by providing a structured digital record of product lifecycle information that can be accessed by multiple stakeholders. Products covered by the DPP assigns them a *Unique Product Identifier* (UPI), which stores lifecycle information such as:

- Origin, model/ID, manufacturer,
- Material composition, hazardous substances, disassembly instructions,
- Environmental footprint, energy efficiency, recycled content,
- Repair history, usage data, warranties.

However, the DPP represents not only a technical infrastructure but also a socio-technical system that must be integrated into existing organizational practices and workflows. Without careful consideration of designing for the user, there is a risk that the DPP explicitly becomes an administrative burden rather than an enabling tool for sustainability. Within the RemaNet project, this thesis addresses the gap between policy-driven ambitions for the DPP and the practical realities of remanufacturing work. Rather than focusing on backend systems or regulatory implementation – like projects such as CIRPASS (2024) – this thesis critically examines how a DPP platform could be designed to support everyday remanufacturing activities in a way that is intuitive and efficient for its users.

1.2 STAKEHOLDERS

The *Digital Product Passport* (DPP) ecosystem involves a wide range of stakeholders across the product life cycle. At a high level, these include manufacturers and suppliers who provide product data, remanufacturers and repairers who rely on this data (for operational decision-making), end users and service providers who interact with the products throughout their use phase, policymakers and regulators who define the DPP framework, and digital platform providers who implement technical solutions. Within the scope of this thesis, the stakeholder focus is deliberately narrowed to enable in-depth exploration, for stakeholder groups were therefore considered most relevant:

- **Original Equipment Manufacturers (OEMs),** primary data providers who supply products specifications and lifecycle information.
- **Remanufacturers,** key data users who depend on accurate and accessible information to assess, disassemble, test, and reassemble products.
- **Technicians,** who interact directly with the DPP platform during remanufacturing operations and represent the primary end users of the proposed platform.
- **Policymakers and regulatory bodies,** who define the requirements and standards that the DPP must comply with.

While the broader DPP ecosystem is acknowledged, the design work in this thesis primarily targets the remanufacturing operator perspective, as this is where the interaction with the platform is most frequent and where usability has the greatest impact on adoption and efficiency on both processing products and logging changes made in their respective DPP's.

1.3 OBJECTIVE AND AIMS

The objective of this thesis is to explore how a *Digital Product Passport* (DPP) platform can be designed to support remanufacturing workflows/processes through a user-centered approach. The work aims to develop usability guidelines, reduce resistance to utilizing the DPP, and a proof-of-concept prototype for the DPP platform by bridging theoretical perspectives on usability, interaction design, and change management with practical design exploration in an industrial context. Specifically, the project aims to:

- Investigate current challenges and adoption barriers related to product data access in remanufacturing contexts.
- Identify how user-centered design principles can be applied to simplify interaction with complex lifecycle data.
- Explore how DPP-related information can be structured and visualized to support efficiency, trust, and acceptance among different stakeholders.
- Develop and evaluate a proof-of-concept prototype that demonstrates how a DPP platform could support remanufacturing operations in practice.

By doing so, the thesis seeks to ensure that the platform based on the DPP system is not only a repository of data but also an efficient and attractive tool that encourages use across the value chain.

1.4 RESEARCH QUESTIONS

The main research question of this thesis is:

- How should the DPP platform be structured, visualized, adapted, and communicated to support user trust, efficiency, and acceptance?

The thesis will also explore the sub questions:

- What adoption barriers and usability challenges do stakeholders face when interacting with the Digital Product Passport system in remanufacturing contexts?
- Which principles of user-centered design and change management can be applied to reduce these barriers and enhance adoption?

1.5 PROJECT SCOPE

This thesis investigates the design of a *Digital Product Passport* (DPP) platform for remanufacturing. The work is limited to concept development and a proof-of-concept prototype, intended to explore how DPP data can be structured, visualized, and interacted with across different stakeholder roles. The scope *primarily* covers front-end design, *including* information architecture, workflow integration, and interaction flows. During the concept phase, three concepts were developed, however only one progressed into an interactive prototype due to time constraints and its relevance to remanufacturing on-site work (the other concepts remain at a conceptual low fidelity level).

Technical implementation aspects such as backend, data infrastructures, cybersecurity, regulatory compliance verification, and integration with existing *Enterprise Resource Planning* (ERP) or *Product Lifecycle Management* (PLM) systems are *excluded*. While existing DPP frameworks and standards are used to inform the design, the thesis *does not* attempt to validate or implement these at a technical or legal level. The user test results gained from the final concept are intended to inform design guidelines and future development rather than provide generalizable performance metrics. The scope is further constrained by the project timeframe and stakeholder access, which limit the depth of iteration and number of user testing during the design process. These limitations are addressed in the discussion as opportunities for future work.

1.6 THESIS OUTLINE

This thesis is structured as follows:

Chapter 1 – Introduction

Introduces the background, context, stakeholders, objectives, research questions, and scope of the thesis.

Chapter 2 – Theoretical Framework

Present the theoretical foundations relevant to the study, including user-centered design, usability, interaction design, technology adoption, change management, circular economy principles, and the Digital Product Passport.

Chapter 3 – Context and Case Description

Describes the industrial and project context of the study, including the RemaNet project and the role of remanufacturing within a circular economy.

Chapter 4 – Method and Implementation

Details the design process and methods used throughout the project, including context immersion, ideation, concept development, prototyping, and user evaluation.

Chapter 5 – Results

Presents the outcomes of the design process, including insights from interviews, user tests, user journey analysis, concept development, and the resulting design proposals and prototype.

Chapter 6 – Discussion

Discusses the results in relation to the research questions and theoretical framework, reflecting on design choices, limitations, implications for future DPP development, and recommendations for future research and design work.

2 Context

This chapter outlines the context in which the Digital Product Passport (DPP) is emerging and situates the thesis within current regulatory and technological developments related to circular economy practices. It provides an overview of the state of DPP initiatives in the European Union, the role remanufacturing within circular value chains, and key challenges with usability, adoption, and data interoperability.

2.1 CURRENT STATE

The shift towards a circular economy is a vital step for a sustainable future according to the European Union, which can be seen being implemented through policies such as the *Ecodesign for Sustainable Products Regulation* (ESPR). The ESPR introduces the *Digital Product Passport* (DPP), which is designed to assign products a unique identifier (UPI), which links them to standardized lifecycle data, such as material composition, maintenance history, and environmental impacts (see Figure 1). The purpose of the DPP is to increase product transparency, extend lifetimes, and to encourage sustainable practices across all sectors (European Parliament & Council, 2024; European Commission, 2025).

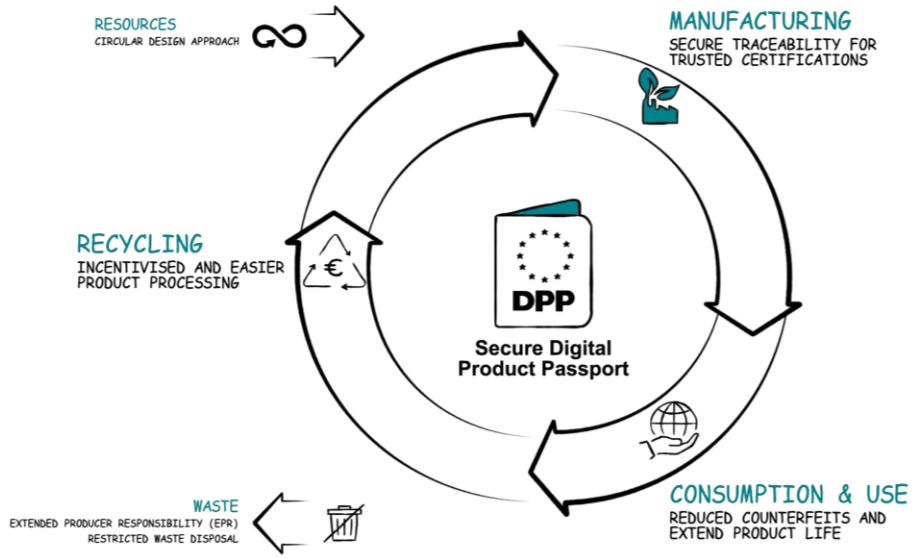


Figure 1. *The DPP visualized, adapted from Certilogo (2023).*

To make the DPP a reality, the EU has initiated several large-scale projects to develop the technical and organizational frameworks required for the DPP. Projects such as CIRPASS and Catena-X are exploring system architectures and interoperability standards for exchanging data between OEMs, manufacturers, suppliers, and end-users (CIRPASS, 2024; Catena-X, 2025). In parallel, projects such as RemaNet, which this thesis situates itself in, address the context of refabrication, embedding the DPP concept into its framework. Through a variety of use cases, RemaNet seeks not only the common goal of tracking environmental impacts such as CO_2 emissions, but also to support safe, efficient, and attractive work environments for remanufacturers (RemaNet, 2024).

Product information is currently fragmented, inconsistent or locked away within organizations. Operators may lack access to vital data for maintenance, while remanufacturers have difficulties finding the source of a specific component within an electric motor. These issues slow down the transition toward a more circular economy (Intereconomics, 2025). There needs to be a platform in place that delivers value for the stakeholders' additional administrative burdens. Through examining how UX and change management principles can promote adoption and minimize barriers to digital system use in remanufacturing contexts – it is these issues that the thesis will address by developing a prototype that demonstrates how DPP data can be transformed from a requirement into a usable and meaningful tool.

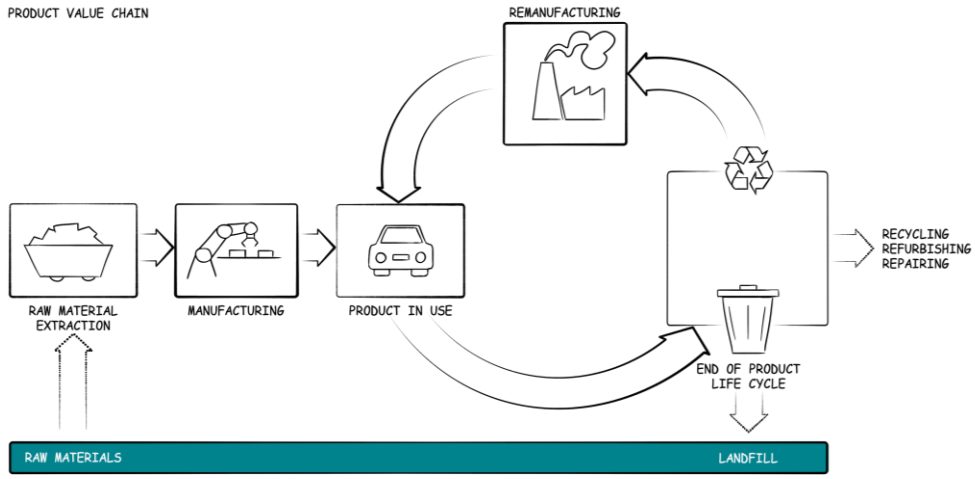


Figure 2. *Simplified product value chain, adapted from McKinsey & Company (2023).*

Conventional manufacturing systems typically have a linear lifecycle for products (raw material to disposal). As seen in Figure 2 (above), circular models, in contrast, aim to close the loop through activities such as reuse, remanufacturing, and recycling (European Commission, 2020). The typical remanufacturing process consists of seven main steps:

Table 2.1.1. *Remanufacturing process.*

Step	Process
Core[1] *Collection*	Retrieval of used products from customers or the field.
Inspection & Sorting	Assessing whether a product can be remanufactured, repaired, or scrapped.
Strip Core (Disassembly)	Systematic disassembly of core to retrieve reusable and remanufacturable components.
Cleaning & Refurbishment	Removing contaminants and restoring components.
Component Testing	Evaluating individual parts for

[1] The part of a product with highest relative value that is being remanufactured is commonly called a *core* (Ijomah et al., 2004).

	compliance with specs.
Reassembly	Rebuilding the product.
Final Testing & Quality Assurance	Ensuring the product performs as intended.

The remanufacturing process not only reduces raw material and energy consumption, but it also offers substantial environmental benefits in comparison to manufacturing. A meta-analysis by Zheng et al. (2019) found that remanufactured mechanical products generally achieve more than a 50% reduction in *Global Warming Potential* (GWP) compared to their newly manufactured counterparts.

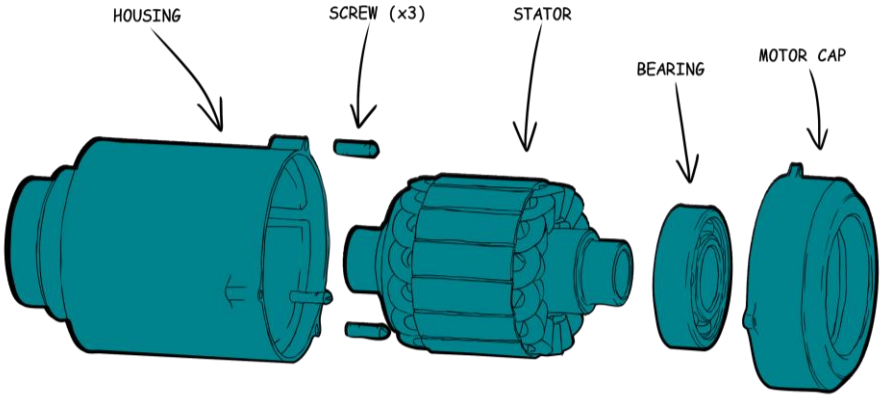


Figure 3. *Simplified DC Motor Explosion View.*

In this thesis, remanufacturing is explored through the case of industrial electric motors – furthermore, the thesis utilizes a DC brushless electric motor with its components clustered into a fewer amount for simplicity (Millett, 2024). An electric motor consists of several high-value components that are particularly relevant for remanufacturing such as the stator, rotor, shaft, housing, and bearings. These parts often retain structural and material integrity after use and therefore allows for remanufacturing with significantly lower resource input than manufacturing new components. Other elements such as worn cabling and seals are typically replaced rather than restored, as their material value and reuse potential are comparatively low. The simplified electric motor used in this thesis (see Figure 3) has five "main" components. The housing consists of the external magnets (which can be seen on the inner surface) and the flange. The stator includes the iron core, winding, and communicator. The electric motor is sealed with the motor cap and three screws, where the wiring, shaft, and brush system is excluded.

Remanufacturing these core components help companies reduce material costs and extend the value of already manufactured products. It also creates opportunities for service-based business models and long-term relationships built around maintenance rather than one-time sales.

2.2CHALLENGES OF USABILITY AND ADOPTION

Despite the ongoing development of the DPP's technical infrastructure, the challenge of making the system usable and adoptable persists. Digital tools often fail not because of the lack of need or flawed technology, but because design principles have been overlooked in the human-machine interaction (Norman, 2013). If a system is cumbersome, unclear, or perceived as adding unnecessary work, the users will resist

it. For the DPP system to fulfil its potential, it must be not only technically sound, but also intuitive and attractive by those who depend on it. Through surveying and interviews across all sectors, CIRPASS (2024) found that stakeholders fear added administrative burden; concerns regarding data confidentiality and interoperability across systems; and that users will require minimal training effort and intuitive UI to engage with new platforms. Understanding these barriers is essential to ensure that relevant stakeholders adopt the platform.

2.3 BENCHMARKING

To better understand the state of emerging DPP ecosystems and identify relevant design principles, as well as areas of improvement and best practices, several platforms and initiatives across industries were benchmarked. These include sector-specific solutions such as the Global Battery Alliance's Battery Passport, cross-sector consortia like Catena-X, blockchain-based applications such as Circularise, and established EU-based databases like EPREL. The table in Appendix A summarizes the findings, comparing sector focus, key features, data-sharing approaches, and governance models.

From this benchmarking, several design and format implications emerge. DPP-based platforms highlight data interoperability, ease of use/access, and stakeholder trust as critical factors for participation. Initiatives also emphasize the importance of clear role-/stakeholder-based data visibility, ensuring that each stakeholder receives only the information relevant to them – in a format they can act on. These insights will influence the design of the proposed DPP platform prototype in areas such as data presentation, login/authorization structures, and integration with the existing workflow of remanufacturers.

2.4 REGULATORY AND POLICY LANDSCAPE

To increase product transparency, reduce environmental impact (throughout the products lifecycle), and enabling circular economy strategies, the European Union's ESPR establishes the legislative foundation for the DPP. The DPP will become mandatory for a range of products groups, such as textiles, furniture, electronics, etc. The DPP will assign each product a UPI, which be linked to standardized data covering environmental footprint, origin, material composition, repair/maintenance history, and end-of-life instructions (European Commission, 2025).

Initiatives such as CIRPASS (and its successor CIRPASS-2) act as complementary to the DPP system, and have been influential in developing technical architecture, data models, and governance frameworks (CIRPASS, 2024). For remanufacturers the regulatory push represents both an opportunity and a challenge – on one hand, standardized access to product data could significantly improve decision-making whether to repair, recycle, or scrap a product. On the other hand, organizations will need to adapt their internal processes, including IT systems and workforce practices, to comply with DPP requirements such as data- and maintenance entry. This duality underscores the need for well-designed, user cantered digital solutions that lower the threshold for participation – which is where this thesis becomes relevant.

2.5 DPP CHALLENGES AND OPPORTUNITIES

While the DPP framework presents an opportunity for advancing circular economy principles – implementing it has its challenges. Semantic interoperability is a key barrier according to CIRPASS (2024), where stakeholders across the value chain must

be able to interpret exchanged data consistently. To not risk useful information becoming fragmented or ambiguous – harmonized taxonomies and data standards must be used across the value chain.

For remanufacturers (and other stakeholders), additional data-entry requirements can be seen as burdensome unless they bring clear operational value – which presents another major challenge – user adoption. CIRPASS *User Stories* revealed that stakeholders desire integration with existing ERP/PLM systems, minimal-effort logging (whilst interacting with a DPP system), and mobile-friendly interfaces for minimal disruption established workflows. RemaNet further highlights these barriers within the context of remanufacturing. Inconsistent or missing product information often leads to delays, higher cost, or unnecessary scrapping of components. By embedding the DPP into a federated digital platform, RemaNet seeks to enhance transparency and traceability throughout the product lifecycle. The improvement of information flow is a key to its broader goal of creating safe, attractive, and efficient work environments for remanufacturers (RemaNet, 2024).

These are not only challenges, but also unique opportunities. A DPP platform with good design can provide real-time product insights, enabling precise environmental impact assessment by reducing decision-making time for remanufacturers. For example, by visualizing CO_2 savings, disassembly instructions, and repair history in a clear and intuitive way, the platform can also strengthen businesses and contribute to EU-wide sustainability goals (European Commission, 2025). For this thesis, these findings underline the importance of a user-cantered approach that not only explores the technical side, but also the psychological and organizational barriers to adoption. Through stakeholder interviews, workshops, and iterative prototyping, this project seeks to transform the challenges into design opportunities – resulting in a platform concept that will promote trust, transparency, and usability.

2.6 PRELIMINARY STAKEHOLDER OVERVIEW

The Digital Product Passport (DPP) ecosystem encompasses a broad network of stakeholders, each playing a distinct role in the creation, exchange, and use of product data throughout the lifecycle. Understanding these relationships is critical for identifying design opportunities and anticipating potential adoption barriers in remanufacturing contexts.

According to CIRPASS (2024) and the European Commission (2025), primary stakeholders include manufacturers, suppliers, remanufacturers, regulators, and end users. Manufacturers and suppliers are responsible for generating and maintaining accurate product information – such as material composition, environmental performance, and disassembly instructions – which form the foundation of the DPP. Regulators and policymakers, including the European Commission and national market surveillance authorities, oversee data governance and ensure compliance with sustainability standards defined under the Ecodesign for Sustainable Products Regulation (ESPR) (European Parliament & Council, 2024).

The DPP ecosystem encompasses a broad network of stakeholders, playing distinct roles in the creation, exchange, and use of product data throughout its lifecycle. To identify design opportunities and adoption barriers, it is critical to understand these relationships. According to CIRPASS (2024) and the European Commission (2025), primary stakeholders include manufacturers, suppliers, remanufacturers, regulators,

and end users.

Table 2.2.1. Overview of key stakeholders in the DPP ecosystem (based on CIRPASS, 2024; European Commission, 2025; RemaNet, 2024)

Stakeholder	Primary Role	Data Responsibility
Manufacturers & Suppliers	Generate DPP and maintain product data (BoM, environmental footprint, disassembly info)	Ensure data accuracy and standardization
Remanufacturers & Repairers	Evaluate, disassemble, and restore products/components	Retrieve product spec, repair histories, material info, performance data, and log events
Regulators & Policymakers	Define and enforce DPP standards and sustainability legislation (ESPR)	Oversee compliance, data governance, and interoperability frameworks
End-users & Service Providers	Use DPP for repair, maintenance, and resale	Access warranties, repair guide, and environmental impact data
Digital Platform Providers & IT Integrators	Develop and maintain DPP infrastructure	Enable secure data exchange and federated access across sectors

This preliminary stakeholder mapping (see Table 2.2.1), based on existing research and policy frameworks, forms the foundation for this thesis. The subsequent interviews with remanufacturers will refine this mapping – revealing specific data needs, workflows, and requirements.

3 Theoretical framework

This chapter covers the theory that guide the thesis. It introduces key points within *Industrial Design Engineering* (IDE) and links them to usability, sustainability, and change management. The aim is to situate the work within established research – to clarify its relevance – and provide the conceptual tools needed to develop a user-centered concept for the DPP in remanufacturing contexts.

3.1 INDUSTRIAL DESIGN ENGINEERING

Industrial Design Engineering (IDE) is an interdisciplinary field connecting design, engineering, and human factors. By balancing technological feasibility, human desirability, and business viability – designers within IDE create products, systems, and services that are meaningful, providing a unique experience to the user (Cross, 2000). Historically, IDE was focused on form and function of mass-produced goods, however, in the current complex socio-technical environment, IDE has evolved to encompass broader systemic and digital challenges (Bourgeois, 2025). Modern industrial designers address user experiences, sustainability, and empathy-driven design. Current processes involve collaborative teamwork within multidisciplinary teams – consisting of engineers, UX/UI designers, strategists, and material scientists – ensuring that solutions are not only functional and feasible, but also meaningful, inclusive, and environmentally responsible (IDSA, 2019).

In this sense, IDE represents a synthesis of creativity and technical rigor, combining what Krippendorff (1989) refers to as a designers "sense-making" ability with a systems-oriented mindset (Buchanan, 2019). The present state of IDE has shifted from designing isolated artifacts to designing interconnected systems, services, and experiences that mediate between people and technology (IxDF, n.d.). For this thesis, a digital platform, this broader perspective is particularly relevant, since the focus extends beyond tangible products to include digital UI, data structures, and collaborative processes.

3.2 THEORY OUTLINE

The theoretical framework is structured as follows:

Usability and User-Centered Design

User-Centered Design (UCD) is an approach that focuses on the needs, behaviours, and goals of the user (Interaction Design Foundation, 2025). Rather than finalizing a product first and then testing it with users, UCD involves users early and continuously – interviews, prototype testing, feedback loops, which Sanders and Stappers (2008) highlight as an effective tool for long-term adoption: *co-creation*. The goal is to create systems that performs currently and at the same time aligns with users' goals, contexts, mental models, and behaviour – essentially having good *usability*. Usability focuses on ensuring that interactions are intuitive, efficient, effective, and satisfying, through varying principles (Norman, 2013; Preece, Rogers, & Sharp, 2015).

Interaction and Emotional Design

A core part of UX design is *Interaction Design*, which concerns how digital systems shape meaningful user experiences through structure, feedback, and transparency (Hallnäs & Redström, 2002; Janlert & Stolterman, 2010). Encompassing the UX field,

is *Emotional Design*, which Norman (2004) describes as how systems look, feel, and behave – which in turn can significantly improve comprehension and trust. Visual clarity, responsive feedback, and an intuitive structure will encourage engagement across stakeholders through the DPP platform.

Change Management and Technology Adoption

Implementing new digital systems also demands organizational change. To establish urgency and embedding new practices – and in turn managing the resistance that arises when introducing new systems – Kotter's (2018) *Eight-Step Model* and Lewins *Change Theory* (Majka, 2024) describe structured processes for guiding transformation. Another way of managing change is explained in Rogers' (1983) *Diffusion of Innovations*, where perceived usefulness and simplicity can influence acceptance rates. Rosson & Carroll's (2002) approach complements this through usability engineering, where they highlight *Scenario-Based Design* (SBD) as a tool to smooth integration into existing workflows.

These concepts form the conceptual basis for this thesis. By addressing the human, organizational, and technological dimensions – they guide the DPP platform's design and evaluation: ensuring a usable, attractive, and appreciated experience within remanufacturing.

3.3 REMANUFACTURING AND THE DIGITAL PRODUCT PASSPORT

Within circular economy, a central strategy is slowing resource loops, or extending product-life, through service loops like remanufacturing (Bocken et al., 2015). The process (see Table 2.1) involves collection, inspection, disassembly, cleaning, replacement or repair of components, reassembly, and testing to meet specifications (Ijomah et al., 2007). Compared to practices such as repair, remanufacturing ensures product performance comparable to a products newly manufactured counterpart – while reducing environmental impacts such as *Global Warming Potential* (GWP) and *Primary Energy Demand* (PED) (Zhang et al., 2019). Remanufacturing, however, still has persistent challenges. Seitz (2007) discusses the lack consistency in core quality and availability, complicating planning and efficiency for the remanufacturer. CIRPASS (2024) highlights the incomplete or missing product data (material composition, repair history, and disassembly instructions) introducing costly delays, which limits scalability.

Organizational and market constraints are also barriers remanufacturers face. Vives (2019) explain how competitiveness is reduced against new manufacturing through tight margins, limited consumer awareness, and weak policy incentives. To address these systematic barriers, technical innovation and data-driven policy is required. The DPP has emerged as a policy-driven response to this range of challenges. The DPP provides a standardized digital record of product lifecycle data to increase transparency, traceability, and circularity across value chains (European Commission, 2025). Access to verified data on material composition/BoM, previous maintenance, and disassembly instructions will significantly improve operational decision-making for remanufacturers. According to CIRPASS (2024) however, overcoming interoperability and adoption barriers is critical for the DPP to provide value rather than only being an administrative burden.

In this thesis, remanufacturing serves as the contextual backbone for exploring where DPP data can reduce inefficiencies and foster more sustainable and resilient

remanufacturing processes.

3.3.1 Circular Economy and the Digital Product Passport

The shift from the traditional linear model of "take-make-dispose" system to the circular economy has created product longevity, resource efficiency, and closed-loop material flows (Bocken et al., 2015). Within the circular economy framework, remanufacturing plays a major role – by extending product lifecycles it reduces raw material dependency and minimizes environmental burdens. The ESPR introduces the DPP to advance the goal of a circular economy, which makes sustainability data accessible throughout the product value chain (European Parliament & Council, 2024). Products will receive a UPI linked to standardized lifecycle information, which will be accessible via digital interfaces through QR codes or RFID tags, facilitating informed decisions across production, use, and remanufacturing.

The DPP is a cornerstone of the EU's Circular Economy Action Plan (CEAP), intended to enhance transparency and stimulate sustainable business models (Intereconomics, 2025). For remanufacturers, it addresses long-standing pain points: missing product data, lack of material traceability, and limited visibility into prior usage (CIRPASS, 2024). However, as CIRPASS (2024) emphasizes, success depends on harmonized data standards, semantic consistency, and user-friendly interfaces. Without these, the DPP risks becoming administratively demanding rather than enabling. In this thesis, the DPP is viewed not merely as a technical system but as a socio-technical innovation. Its usability, interoperability, and adoption will determine whether it becomes an enabler of sustainable circular practices or an additional layer of complexity in industrial operations.

The DPP is a cornerstone of the EU's Circular Economy Action Plan, intended to enhance transparency, boost global competitiveness, and stimulate sustainable business models (Intereconomics, 2025). Within the context of remanufacturing, the DPP addresses pain points such as missing product data, lack of material traceability, and limited knowledge of previous usage (CIRPASS, 2024). The DPP is not only viewed as a technical system, but a socio-technical innovation; and through platforms such as the proposed proof-of-concept, will become an enabler of sustainable circular practices (and not an additional layer of complexity in industrial operations).

3.4 USER-CENTERED DESIGN

UCD provides a structured approach to ensure systems meet real user needs, and in the case of the DPP platform, ensures that it's efficient, intuitive and relevant to its users. Usability, an element within UCD, refers to the extent which a system enables users to achieve specified goals effectively, efficiently, and satisfactorily (Norman, 2013). In this context, even though the underlying technology is technically sound, poor usability can significantly undermine the goal of adoption. UCD ensures the success of the DPP platform, through iterative development, stakeholder participation, and context-based evaluation to align system with actual work practices (Preece, Rogers, & Sharp, 2015; Sanders & Stappers, 2008). Adapting this process means involving operators, engineers, and managers to capture diverse requirements ranging from shop-floor data logging to lifecycle reporting for sustainability metrics.

Prototyping and *wireframing* play an important role in translating user insights into testable design concepts. Iterative testing of low- and high-fidelity prototypes (Wikberg Nilsson et al., 2017) enables validation of design assumptions and will also reduce the risk of low adoption rates when the system is implemented. Norman

(2013) describes key theories within usability, such as *affordances* and *signifiers* – which, respectively, describe how users perceive possible actions and cues about how to perform them. Clear visual signifiers in a DPP platform are important to guide user interaction and to minimize cognitive load. Similarly, Nielen's (1994) *ten usability heuristics* remain widely applied in system evaluation, for assessing and improving the user experience – visibility, mapping, freedom, consistency, error prevention, recognition, flexibility, minimalism, help, and error recovery. In a DPP platform context these heuristics translate into features like immediate confirmation after data entry, consistent interface patterns across devices, and terminology familiar across all industries.

3.5 INTERACTION, EMOTION, AND BEHAVIOUR IN DESIGN

3.5.1 Interaction Design

Within *UX* and *Human-Computer Interaction* (HCI) lies *Interaction Design* (IxD) – which concerns how digital systems are structured to support meaningful and efficient exchanges between people and technology (Interaction Design Foundation, 2025). Beyond functionality, IxD addresses how a system communicates through for instance signifiers (which optimize affordances), feedback, navigation, and temporal flow. Through these factors, users can confidently perceive and execute actions – and understand the outcomes that follows (Hallnäs & Redström, 2002; Norman, 2013). In the context of the DPP platform, IxD decisions determine whether a stakeholder using the platform can swiftly retrieve, interpret, and act on lifecycle data.

Federated DPP architectures introduce what Janlert and Stolterman (2010) term *interaction complexity*, which is where multiple layers of objects, states, and relations can easily exceed human working-memory limits. To manage these complexities, it is required to have disciplined information and flow design. Core strategies include:

- **Hierarchical organizational and wayfinding:** clear primary navigation (What domain am I in: product, component, event?) with reliable local navigation (What can I do here?).
- **Progressive disclosure:** reveal detail as needed, keeping default views concise (Materials can for example be expanded to hazardous substances).
- **Task-first scaffolding:** guide users via action pathways (Scan -> Identify -> Act).
- **State visibility:** immediate confirmations, non-blocking toasts[2], persistent status badges.
- **Error prevention and graceful recovery:** constrained inputs, inline validation, undo, and clear provenance ("Source: OEM, Timestamp").
- **Consistency and learnability:** uniform patterns across devices, predictable label language (aligned with shop-floor vocabulary), and stable placement of recurring controls.

These strategies translate into concrete IxD choices for the DPP platform:

- Scan-first entry point through QR/RFID.
- Split-view layout balancing specs and tasks.
- Filterable tables.
- Drill-down panels for provenance, service history, and hazard flags.
- Roll-up dashboards for managers and policymakers (CO_2 savings, etc.).
- Role-literate and context-literate content
- Minimize navigation and input for time-pressed users

Collectively, these principles reduce cognitive friction, shorten time-to-insight[3], and build the conditions for trust and sustained use.

[2] Notifications that appear, for example in the corner of a GUI, are referred to as a *toast* (Microsoft, 2025).
[3] The duration it takes to convert data into actionable insight that enables a decision (Holt-Nguyen, 2023).

3.5.2 Human–Machine Interaction and Emotional Design

Human-Machine Interaction (HMI), acts as the bridge between humans and complex technologies, examining how people perceive, act through, and make sense of interactive systems. By integrating *ergonomics*, *cognition*, *perception* and *affect*, HMI enables effective and meaningful interactions (Pandey & Nathani, 2023). In the current industry, HMI draws on multimodality, utilizing visuals, tactility, and audio, to adapt interfaces to context. Safety, reliability, and clarity are first-order qualities in industrial HMI and should be explicitly designed into every feedback loop. Modern HMI also recognizes that usability when using a product not only is cognitive, but also emotional. Norman defines the *three levels of emotional processing*; visceral (first impression); behavioural (use-in-action); and reflective (after-use meaning); which show how aesthetics, rhythm, and narrative shape trust and willingness to return (Norman, 2004). Emotional design is valuable in the context of a data-heavy tool like the DPP platform.

Adjacent fields reinforce this stance – there's *affective computing* which explores how systems sense and responds to human emotion (Picard, 1997), and *kansei/affective engineering* which investigates mapping human feelings to product attributes (Asada, 2014). These frameworks motivate interface choices that feel dependable, for example, consistent typography, stable layouts, and clear explanations for system decisions. In federated data settings like the DPP platform, explainability and transparency are very valuable, since users must equally know what a system says, and why it is says it. Insights from Japanese social robotics further illuminate how interfaces can invite cooperation. The concept of "weak robots" introduced by Okada (2023) shows that revealing limitations can elicit user support. This becomes honest UX when translated to interfaces – when data is missing or uncertain, the system states so plainly ("No OEM service record found"), and offers next-best actions ("Suggest: request from distributor"). By doing this, the interface doesn't mask gaps, rather it acknowledges them and helps users resolve the issue. Paired with provenance cues, this stance will support trust in the DPP platform, where partial data are common. HMI in this context ultimately should:

- Prioritize readability and speed-to-action in noisy or high tempo contexts.
- Communicate certainty and uncertainty when it comes to provenance, timestamps, etc.
- Use emotion as a way for clarity: calm feedback, confident micro-copy, coherent visual rhythm.
- Adapt to role and device, keeping mental models.

3.5.3 UX and Behaviour Design

To design for adoption requires understanding not only how users interact with systems, but also why they choose to engage/disengage with them. Behavioural and cognitive psychology is of up most importance when it comes to shaping the UX of the DPP platform, aligning usability with human motivation.

Cognitive Load Theory (Sweller, 1988) highlights that human memory is limited, therefore overload can occur when interfaces present excessive or poorly organized information – this can lead to frustrations or avoidance. In remanufacturing contexts, operators already face cognitively demanding decisions (identifying cores, assessment, etc.) which is why the DPP platform must reduce extra loads. Methods like progressive disclosure, visual hierarchy, and task-driven workflows are therefore important to utilize, ensuring users see only what is relevant at each moment.

Fogg's Behaviour Model (Fogg, 2009) explains that when motivation, ability, and

triggers coincide; behaviour occurs. In the context of the DPP platform, even though an operator understands the DPP's value, they will not engage if tasks feel too complex (low ability) or lack strong cues (low triggers). The GUI of the DPP platform must therefore balance effort and motivation, where actions should require minimal effort while clear feedback provides a motivational cue. System notifications can act as triggers, for instance to complete missing fields, without feeling intrusive.

Nudge Theory (Thaler & Sunstein, 2008) complements this by suggesting that users can be guided toward desirable behaviours (while maintaining autonomy), through subtle design interventions. Nudges can make sustainable and compliant behaviour the path of least resistance, and through that embeds environmental responsibility into everyday workflows without explicit enforcement. In the context of the DPP platform, nudges can be:

- Default selections that match most sustainable or most likely to be selected within options.
- Progress indicators.
- Contextual prompts.
- Visible sustainability metrics.

3.5.4 Evaluating Behaviour and UX

To ensure that behavioural insights translate into effective design, usability evaluation must be continuous and based on evidence. *Heuristic evaluation* (Nielsen, 1994) is performed to allow experts to assess prototypes against the core usability principles. *Cognitive walkthroughs* (Wharton et al., 1994) can stimulate first-time user experiences, which can reveal whether new users intuitively can complete critical tasks, such as scanning a QR code in the DPP platform, without prior training. Nielsen also presses on the importance of test subjects verbalizing their thoughts during an evaluation, referred to as a *Thinking Aloud*, which is a robust tool for assessing users initial understanding of an interface (Nielsen, 2012). *Usability testing* (Rubin & Chisnell, 2008) is done later in the design process, through engaging real stakeholders in hands-on scenarios. Through observation of user behaviour evidence for refinement is collected. Quantitative metrics such as task completion rates, error frequency, and the *System Usability Scale* (SUS) score (Soegaard, 2023) supplements the qualitative insights. The iterative feedback loop, as seen in the *Double Diamond Design Process* (Design Council, 2007), ensures that the DPP platform evolves with its users.

Interaction Design, HMI, Behavioural Design, and User Experience converge on a shared goal: creating systems that are intelligible, trustworthy, engaging, and that provides value and satisfaction to the user. These are ultimately not tools for aesthetic luxuries, but essential conditions for the adoption of circular economy technologies. A well-designed DPP platform interface should therefore:

- Present information with clarity and honesty.
- Support effortless, low-friction interaction, that is aligned with the users' behavioural drivers.
- Offer feedback that builds trust and reflects environmental value.
- Encourage consistent participation through subtle motivation and usability cues.

3.6 CHANGE MANAGEMENT & TECHNOLOGY ADOPTION

New systems like the DPP are not simply a technical deployment, it is an organizational and cultural transformation, which will disrupt existing routines, redistribute responsibilities, and may challenge assumptions about workflows and data ownership. Change management is therefore a crucial theoretical foundation for ensuring that digital innovation translates into sustainable practice, rather than a short-

lived experiment.

3.6.1 Organizational Change Frameworks

Kurt Lewin's (1947) *Three-Stage Model* conceptualizes change as a cycle of unfreezing, changing, and refreezing. During the unfreezing phase organizations will initially create awareness of the need for change, for instance by surfacing inefficiencies and regulatory pressures. The changing phase involves introducing new processes such as systems or behaviours, through iterative adaptation, experimentation, and training. The refreezing phase combines the new practices into everyday operations and therefore embeds them within the organizational norms and culture. In the context of the DPP platform, this sequence can be seen when a remanufacturing firm recognizes inefficiencies when tracking product lifecycles, identifying a product, or assessing whether a core can be remanufactured (unfreezing). A digital platform is therefore piloted to help with these concerns (changing), which is then integrated into the operation as a standard (refreezing).

John Kotter's (2018) *8-Step Model for Leading Change* extends Lewin's framework into a detailed roadmap which focuses on leadership and communication. The steps consist of creating a sense of urgency; building a guiding coalition (of leaders); form a strategic vision; communicate that vision; enlist a volunteer army; generate short-term wins; sustain acceleration (continue to build on the momentum); and finally, institute change. These steps stress the need for alignment across hierarchies where leadership must model enthusiasm for the DPP's sustainability goals, while mid-level managers ensure that operators see clear benefits in their daily work.

Both models converge on a key insight: resistance to change is not inherently negative. Resistance is a natural response to uncertainty or perceived loss of control. Addressing resistance requires transparency, participation, and early wins. Through the DPP platform, demonstrating time savings or error reduction is a satisfactory way of doing this.

3.6.2 Technology Adoption Theories

While change management focuses on organizational readiness for change, technology adoption explains how and why individuals decide to use new systems. *The Diffusion of Innovations* framework (Rogers, 1983) identifies five attributes that influence adoption rates:

- **Relative advantage:** the perceived improvement over current methods.
- **Compatibility:** alignment with existing workflows, systems, and values.
- **Complexity:** the perceived ease of use or difficulty.
- **Trialability:** opportunities to test the system on a small scale.
- **Observability:** visibility of positive results to others.

For the proposed DPP platform and the overall DPP system, this means that stakeholders must see clear advantages (faster data retrieval, regulatory compliance, reduced waste) while maintaining workflow continuity with ERP/PLM systems. Complexity should be minimized through intuitive interfaces and minimal manual data entry, while pilot implementations (which in this context will be user testing of prototypes) can enhance trialability and observability. In industrial settings where productivity pressures are high, perceived effort can outweigh function benefits, which according to Norman (2013) can lead to users abandoning powerful tools (due to the friction in interaction). The CIRPASS (2024) consultation findings mirror this: industrial stakeholders consistently prioritize simplicity, interoperability, and

minimal-effort interfaces (for instance, mobile scanning), over advanced but complex features.

Change management and adoption theory therefore converge on the fact that technology should be designed and implemented in a way that fits people's reality.

3.6.3 Embedding Change Through Leadership and Participation.

For digital transformation initiatives like the DPP, leadership plays a decisive role. Kotter's (2018) model underscores the importance of visible leadership engagement, where leaders who actively use, endorse and reward system use will set the tone for organizational culture. Equally vital is participatory change, according to Sanders and Stappers (2008), where co-creation and participatory design can foster ownership by allowing stakeholders to contribute to the design of the tools they will use. Early involvement in prototyping or workflow mapping helps ensure that the DPP platform reflects real operational constraints rather than idealized/theoretical models. Simultaneously, this participatory process supports Lewin's (1947) Three-Stage Model, as stakeholders see their feedback materialize in design iterations.

3.6.4 Feedback Loops and Continuous Adoption

Although this thesis hopes to provide proof-of-concept and guidelines for implementing a DPP platform within remanufacturing contexts, it is vital within change management for ongoing evaluation and feedback loops post-deployment. Feedback mechanisms such as in-app prompts informing improvement or dashboards showing environmental benefits can transform abstract sustainability targets into visible and motivating evidence for the user. These small but continuous reinforcements enable what Rogers (1983) calls observability, strengthening the long-term commitment to the DPP platform. Additionally, adaptive governance, which consists of periodic reviews of workflows, feedback sessions from stakeholders, and design updates, keeps the DPP system responsive to evolving industrial and policy conditions. By framing technology adoption as an iterative and co-evolving process rather than a one-time rollout, organizations can accommodate regulatory updates, new standards, or shifts in user priorities without wearing down trust.

3.7INTEROPERABILITY AND DATA STANDARDS

Interoperability forms the technical and structural foundation of the Digital Product Passport (DPP). It ensures that information exchanged between stakeholders– manufacturers, remanufacturers, regulators, and end users– can be shared, understood, and reused consistently across systems and sectors. In this sense, interoperability extends beyond data exchange to encompass semantic alignment: not only must systems communicate, but they must agree on what their data means (CIRPASS, 2024).

Achieving this requires the use of standardized data models and ontologies. International bodies such as ISO, IEC, and GS1 provide frameworks for identifiers, product classification, and data exchange that promote global consistency. Within the European Union, the DPP must also comply with the Ecodesign for Sustainable Products Regulation (ESPR) and draw upon related databases such as EPREL, which already manages energy labelling data for electrical products (European Parliament & Council, 2024). These initiatives illustrate how harmonized standards can link material, product, and process information throughout a product's lifecycle.

For remanufacturing, interoperability is particularly crucial. Remanufacturers often

handle components from multiple original equipment manufacturers (OEMs), each with their own documentation formats. Without common standards, data incompatibility leads to inefficiency, duplication of effort, and costly delays. Conversely, a unified approach– using interoperable identifiers, standardized metadata, and open exchange protocols– enables traceability, simplifies lifecycle management, and supports scalability across industries.

However, interoperability is not solely a technical concern; it is also a human-centered design challenge. Complex data architectures must be made accessible and actionable for non-expert users. Therefore, the design of the DPP interface must balance back-end harmonization with front-end usability– ensuring that standardized information structures translate into clear, meaningful, and intuitive user experiences. This thesis explores that intersection, positioning interoperability not just as a regulatory requirement but as a driver of transparency, efficiency, and collaboration within circular value networks.

3.8 DESIGN RESEARCH AND GUIDELINES

Design Research provides a structured framework for exploring how technology interacts together with people and organizations, which is an essential foundation for developing a DPP that is both usable and adoptable. Strategies such as traditional engineering methods focus on mostly functionality, design research however, integrate human-centered examination, iteration and reflection to address complex, real-world problems (Blessing & Chakrabarti, 2009).

3.8.1 Design Research Methodologies

The *Design Research Methodology* (DRM) by Blessing and Chakrabarti (2009) divides research into four iterative stages:

- **Research Clarification:** defining objectives, identifying gaps, and understanding contextual constraints.
- **Descriptive Study I:** exploring current practices and user challenges through observation and data gathering.
- **Prescriptive Study I:** developing conceptual models, frameworks, or design interventions.
- **Descriptive Study II:** evaluating the effectiveness and applicability of proposed solutions in practice.

This structure aligns closely with the iterative development of the DPP proof-of-concept platform in this thesis, moving from understanding remanufacturing workflows to formulating design principles and evaluating their effectiveness through stakeholder engagement. Complimenting DRM, *Needfinding* (Patnaik, 2017; Björklund & van der Marel, 2025) can be applied to uncover underlying user needs that may not surface in traditional interviews or surveys – this is done by designers immersing themselves in real work environments. Inefficiencies, emotional pain points, and behavioural patterns can be identified, informing more meaningful solutions. This goes together with co-creation (Sanders & Stappers, 2008) where stakeholders are positioned as active contributors in the design process.

The Double Diamond Design Process (Design Council, 2007) further guides the thesis through four key phases, which supports a balance between divergent and convergent thinking:

- **Discover**: gather knowledge through literature, benchmarking, and interviews.
- **Define:** analyse insights, formulating user requirements and adoption guidelines.
- **Develop:** generate ideas, build wireframes, and iterate prototypes.
- **Deliver:** refine the proof-of-concept, produce usability guidelines, and communicate results.

Together, these methods create a methodological backbone that emphasizes iteration and collaboration together with reflection, which ensures that the design process remains both systematic and human centered.

3.8.2 From Methodology to Design Principles

Design research also translates findings into design principles, which guide the creation of systems that are not only usable, but also inclusive, and efficient. Within this thesis, usability theory and interaction principles (Norman, 2013; Nielsen, 1994; Jordan, 1998) form the foundation of these guidelines which ensures that the design decisions are empirically grounded, and user focused.

A major concern within the DPP platform is to minimize cognitive and physical effort, which Nielsen (1994) stresses can be done by reducing unnecessary steps and complexity on interfaces. Norman (2013) introduces the idea of natural mapping, where designers can design controls and layouts that correspond logically to users' mental models, which in the context of remanufacturing and the DPP platform would be aligning the navigation with the physical workflow.

As Jordan (1998) notes, users rely on familiar patterns to navigate systems efficiently. To support learnability and prevent confusion, *consistency* should be applied both internally (within the DPP platform) and externally (with other tools like ERP/PLM systems). Factors like error prevention and recovery is equally critical for the DPP platform, which is emphasized by Norman (2013) and Nielsen (1994). Mechanisms like confirmation prompts and undo functions therefore help users maintain confidence and trust.

Flexibility and efficiency of use (Shneiderman & Plaisant, 2005) are equally vital where interfaces should accommodate both novice users through guided actions and expert users through shortcuts or automation. Finally, a central role to usability is feedback, where users must receive clear and immediate signals that their actions were successful or that other input is required (Norman, 2013) – this could include visual cues confirming successful data upload. Collectively, these principles define a user experience that is transparent and reliable.

3.8.3 Applying Design Principles in Practice

Design research becomes meaningful only when translated into applicable design guidelines that directly instruct how the platform will be built and experienced. The following guidelines synthesize the covered principles to ensure the platform is practical and aligned with stakeholder needs.

Mobile vs. Desktop

Remanufacturing environments are physically demanding and often involve time sensitive operations, which is why operators may need quick and hands-free access to information. Managers or sustainability analysts, however, work with detailed datasets or reports in office environments. To provide accessibility across the value chain, designing an adaptive interface that functions across platforms (mobile, desktop, tablet) is of merit (Preece, Sharp, & Rogers, 2015). A way of supporting this approach is through a mobile-first approach, where simplified navigation paired with high-contrast interactive elements is of priority. This would enable QR code scanning, simple guides, real-time logging, and digital checklists – which would also reduce paperwork. Through desktop interfaces meanwhile, advanced features such as

analytical dashboards and reporting tools can be offered.

Feedback vs. Transparency

To ensure accuracy in complex data environments, as well maintaining trust from the user, immediate feedback and system transparency is important. Nielsen (1993) presses on the reality that users always should know the system's status, whether it's if a file that has been uploaded or data being saved. In the context of the DPP this would be instant confirmation messages or progress indicators, which can help reduce uncertainty and reduce cognitive strain. Visual feedback loops such as dynamic indicators showing carbon footprint savings would extend the function of feedback beyond reassurance, becoming motivational tools that reinforce goals (Thaler & Sunstein, 2008). Transparency in data history, for instance when a data point was last updated by an OEM, would also strengthen accountability and traceability – key requirements for remanufacturing.

Accessibility and Inclusivity

Inclusive design, as defined by Nielsen Norman Group, focuses on creating products that understand and enable people from all backgrounds and abilities (Kendrick, 2022). Throughout the DPP ecosystem, stakeholders will have a varying level of digital literacy, technical expertise, physical context, and linguistic background. Designing for inclusiveness within that aspect ensures that every user can engage with the system in a satisfactory and effective way. Interfaces should prioritize clear and interoperable language (cultural variation), consistent iconography, and typography that is appropriate. Within the context of the DPP platform – it's important to design for users in a noisy environment, with poorly lit conditions (situational challenge) – as well as technicians with limited time and variable connectivity (economic/operational constraint). Customizable or role-based dashboards are a key design choice here, where an operator for example may prioritize disassembly instructions instead of a general view of emissions data, which would be more appropriate for an environmental manager. Factors like multilingual support are equally essential in multinational value chains, ensuring no stakeholder is excluded due to language barriers (Jordan, 2000). Inclusive design also enhances resilience, since when systems are easy to learn and navigate, they will remain functional even under pressure or staff turnover. This will reduce the dependence on specialized training for the platform, ensuring scalability across organizations and geographies.

Design Strategy

The guidelines above illustrate that the usability of the platform is not a secondary layer of design but a strategic enabler of adoption. An intuitive and transparent DPP platform enhances both operational efficiency and stakeholder trust, aligning human behaviour with sustainability objectives. Ultimately, effective design research ensures the DPP platform moves beyond compliance and becomes a genuinely empowering tool.

4 Method and Implementation

This chapter describes the practical execution of the thesis work to provide transparency regarding what was done, how it was done, when different methods were applied, and why they were appropriate for addressing the research questions. The methods employed are rooted in Industrial Design Engineering (IDE) where iterative exploration, prototyping, and reflection are central. The chapter focuses on how selected methods were applied within this project context and will allow the reader to understand how theoretical perspectives, empirical insights, and design activities informed the development of the proof-of-concept prototype.

4.1 PROCESS

This thesis followed an adapted Double Diamond design process (see Figure 3) which consists of four main phases: Discover, Define, Develop, and Deliver (Design Council, 2007). This process was selected because it is well suited for exploratory design challenges where both problem understanding and solution development must progress iteratively. The Double Diamond explicitly separates divergent and convergent phases which suits the broad exploration of the DPP context and the focused development of design concepts.

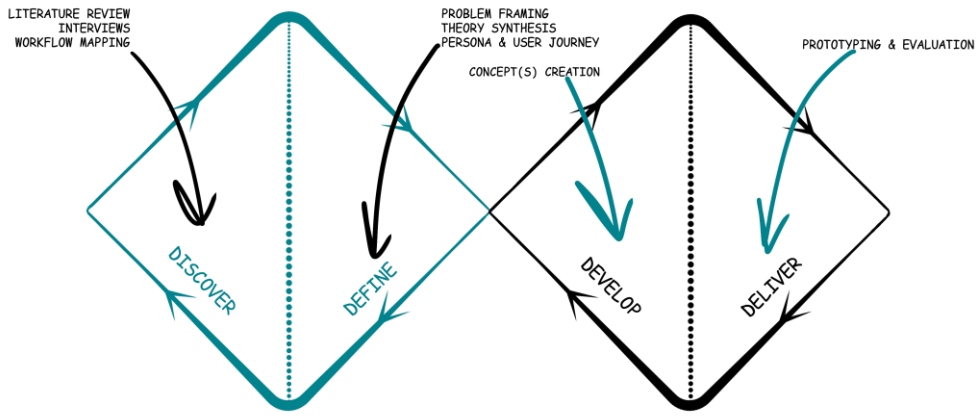


Figure 4. *Adapted Double Diamond design process (Design Council, 2007).*

- **Discover** involved literature review, stakeholder mapping, and context immersion to understand remanufacturing workflows, data challenges, and adoption barriers.
- **Define** focused on synthesizing insights into a structured user journey, identifying key pain points and corresponding opportunities, and framing design directions.
- **Develop** consisted of ideation, sketching, wireframing, prototyping of alternative concepts, and doing user testing.
- **Deliver** focuses on refining a solution into an interactive proof-of-concept prototype.

The process was iterative, and insights from later stages repeatedly informed earlier decisions, particularly during prototyping, where wireframes and interaction flows were revised based on UX considerations and continuous reflection.

4.2 PROJECT PLANNING

Project planning was guided by the project plan approved at the start of the thesis,

where a Gantt chart (see Appendix B) was used to structure the project timeline and align activities with key milestones, including the project proposal, midterm seminar, user testing, and final submission. The Gantt chart included the Double Diamond phases to ensure that sufficient time was allocated for early exploration and later refinement. The thesis stage gates are structured to align with typical design research milestones (Blessing & Chakrabarti, 2009), ensuring that each phase produces validated outputs. Figure 4 shows the project divided into phases where tasks and expected outcomes are assigned for each respective phase.

	Phase	Title	Week	Date
	1	Exploration & Planning	36–39	Sep 1 – Sep 28, 2025
Task	Literature study, benchmarking, stakeholder mapping, userstudy consent preparation.			
Outcome	Research base, interview guide, and project plan submitted.			
	2	User Research & Insights	40–43	Sep 29 – Oct 26, 2025
Task	Conduct stakeholder interviews/workshops. Analyze findings and extract user insights & requirements.			
Outcome	Mapped user needs and draft requirements specification.			
	3	Concept Development & Early Prototyping	44–47	Oct 27 – Nov 23, 2025
Task	Develop concepts and Figma wireframes. Build first prototype (v. 1) and prepare for usability testing.			
Outcome	Preliminary concepts and interactive wireframes.			
	4	Iterative Prototyping & Testing	48–52	Nov 24 – Dec 31, 2025
Task	Test prototype v1, refine into prototype v. 2. Conduct second usability test, refine into prototype v. 3.			
Outcome	Improved proof-of-concept prototype.			
	5	Reporting & Deliverables	36–52	Sep – Dec 2025
Task	Ongoing report writing: Introduction, Context, Theory, and Methods.			
Outcome	Half-time thesis submission and midterm presentation (Week 42–43)			
	6	Finalization	1–5	Jan 1 – Jan 30, 2026
Task	Complete Results, Discussion, and final report chapters. Prepare final presentation, poster, and 2–3 min video. Submit 90% and 95% drafts for supervisor and peer review. Incorporate feedback, finalize report, and submit for approval.			
Outcome	Final seminar presentation (Week 3), poster, video, and final approval (Week 5).			

Figure 5. *Phases.*

Regular check-ins with the academic supervisor and both formal and informal discussions were done with RemaNet representatives to validate direction and scope, particularly when prioritizing which concepts should move forward into detailed prototyping.

4.3 CONTEXT IMMERSION

4.3.1 Interviews

To explore current practices and challenges surrounding product data and remanufacturing workflows, a semi-structured interview was conducted with a representative from Wexoe. The participant was selected based on professional experience in sales, competence, transformation, and operations of remanufactured industrial components, providing insight into both customer-facing process and internal remanufacturing workflows. The interview followed a semi-structured

format to balance consistency with flexibility, allowing deeper exploration of topics that emerged during discussion (Wikberg Nilsson et al., 2017). Questions were derived from the theoretical framework, and from the context immersion – UX and technology adoption played a big role in forming the questions to ensure alignment between empirical findings and design theory.

Five thematic areas guided the interview:

- **Current Practices and Pain Points:** understanding how remanufacturing currently operates, what data is currently available, and where inefficiencies occur.
- **Data Needs:** identifying what product and lifecycle information would be most valuable in a digital format.
- **Platform and Access Preferences:** exploring how users would prefer to interact with a DPP platform and what devices/interfaces are commonly used.
- **UX and Adoption:** examining motivation, ease of use, potential barriers, and discussing needs for everyday adoption.
- **Interoperability and Scaling:** discussing multi-OEM data challenges and requirements for system compatibility.

Example guiding questions included:

- "How does your current remanufacturing process work today?"
- "What kind of product information do you typically receive with incoming products?"
- "What data points would be most helpful to have digitally available?"
- "What would make you more likely to use a DPP platform daily?"
- "How important is it that the platform works for multiple OEMs in the same interface?"

The interview lasted approximately 40 minutes and was conducted via Microsoft Teams. The session was recorded with the participants consent and later transcribed for thematic analysis. Answers were anonymized to ensure confidentiality and ethical compliance. The complete interview guide, including all questions, is provided in Appendix C.

4.4 LITERATURE REVIEW

A literature review was done for the design and evaluation of a DPP platform that supports remanufacturing, ensuring the DPP is attractive within the circular economy. The review aimed to identify frameworks, models, theories, and empirical studies addressing usability and UX, HMI/HCI, change management, and digital transformation in industrial contexts.

Searches were conducted between September and October 2025 using databases and search engines including *Scopus, Google Scholar, ScienceDirect, SpringerLink, and the Luleå University of Technology Library search portal*. Both academic and policy-oriented sources were considered to reflect the interdisciplinary nature of the topic. However, information regarding design from well-established institutions such as the *Interaction Design Foundation*, were presumed reliable albeit not being directly stated as peer reviewed.

4.4.1 Search Terms and Strategy

Search strings combined key terms through Boolean operators (AND, OR) or (+, -) across several conceptual clusters, for example:

- "User-centered design" OR "UCD" AND "usability" AND "industrial design engineering"
- "Interaction design" OR "emotional design" OR "human-machine interaction" "-physical artifact"
- "Digital Product Passport" + "circular economy" + "remanufacturing"

- "Technology adoption" OR "change management" OR "diffusion of innovation"
- "interoperability" AND "data standards" OR "product lifecycle management"

4.4.2 Selection and Inclusion Criteria

Sources were included if they:

- Addressed usability or design methodology in industrial or digital environments.
- Discussed frameworks for technology adoption or change management.
- Related to remanufacturing or circular economy strategies.
- Were written in English and published between 200-2025.

Although the main range was 2000-2025, several foundational works predating 2000 were included, such as Kotter (1996), Rogers (1983), and Sweller (1988), due to their lasting influence on contemporary theories of design and organizational change. Relevant course literature from Luleå University of Technology, such as works from Wikberg Nilsson et al., (2017), were exempt from the English-only-constraint.

4.4.3 Literature Review Outcome

The initial search identified approximately 120 publications, which were screened for relevance and methodological rigor. After filtering duplicates and off-topic items there was around 60 core sources that were retained. These were grouped thematically to align with the theoretical framework. This structured approach ensured both breadth and depth in coverage, allowing theoretical perspectives to be triangulated across disciplines that are relevant to Industrial Design Engineering, Change Management, and the RemaNet project (encapsulated by the DPP).

4.5 IDEATION

The ideation phase focused on generating design concepts, where the objective was to explore a wide solution space before gradually narrowing it down into three distinct concepts (A, B, and C), each addressing different stakeholder perspectives within the DPP ecosystem. The ideation process followed an iterative and divergent to convergent approach, aligned with the Double Diamond process.

4.5.1 Idea Generation

Ideation began with a tweaked version of *braindrawing*, an alternative to traditional brainstorming, where ideas are visually sketched out rather than verbally articulated. Braindrawing was chosen to encourage rapid, playful exploration and to lower the threshold for expressing incomplete or unconventional ideas (Wikberg Nilsson et al., 2017), where notes were added to drawings to understand the ideas further. This stage was intentionally focused on quantity over quality, allowing any idea to emerge without immediate evaluation. These early sketches included unrealistic or "silly" ideas, which served as catalysts for later refinement rather than final solutions. The sessions explored questions such as:

- What could a DPP platform be if no technical or organizational constraints existed?
- How might different stakeholders experience the same product data differently?
- How might you present dense product information to a user?

Given the conceptual complexity of DPP, circular economy frameworks, interoperability standards, and understanding the whole picture, *braindraining* was used as a complementary method. Braindraining as described by Wikberg Nilsson et al. (2017), involves temporarily stepping away from active problem-solving through activities such as walking or meditating reflection. In this project braindraining was used between ideation sessions to prevent cognitive fixation and allow subconscious processing of complex system relationships. Insights gained during these breaks were

subsequently translated into sketches and notes, which were documented in a Miro board, often leading to alternative framings of the problem.

4.5.2 Filtering Ideas

Following the divergent ideation phase, generated ideas were systematically filtered and grouped based on relevance, feasibility, and alignment with the thesis objectives, where the principle *"kill your darlings"* was actively applied, removing ideas with high personal appeal but low relevance (Wikberg Nilsson et al., 2017). The filtering process resulted in three primary concepts:

- **Concept A:** An efficiency-focused, operator-facing DPP platform that supports the remanufacturing workflow on the shop floor.
- **Concept B:** A managerial view of DPP data, emphasizing sustainability metrics, compliance, decision support, PLM data, and standard ERP data including KPIs and trends.
- **Concept C:** A costumer-facing perspective, where tracking a products remanufacturing process was the focus, enabling transparency and feedback throughout the process.

To better emphasize and understand frustrations of the concept's users – a persona (Wikberg Nilsson et al., 2017) was created for each concept based on their corresponding stakeholder, together with a workflow. Supplementing the workflow of Concept A, a user journey (Wikberg Nilsson et al., 2017) was also created to point out pain points (extracted from the context immersion) where a DPP-enabled platform could provide value. The user journey 'touchpoints' were directly correlated to each stage in remanufacturers workflow.

To manage the inherent complexity of the DPP domain the design principle of *simplicity*, the method referred to as *Occam's Razor*, was applied throughout the ideation, where the focus wasn't about removing content but rather reducing the non-essential while preserving what is meaningful (Wikberg Nilsson et al., 2017). The reason for this tie back to user-centered design, where systems such as the DPP risks overwhelming users if its information is presented without prioritization. This principle guided decisions such as:

- Reducing visible information to task-relevant data depending on user role,
- Structuring interfaces to support learnability and efficiency,
- Hiding advanced information until it becomes contextually relevant.

4.6 IMPLEMENTATION

The implementation phase focused on translating the outcomes from the ideation and definition of stages into concrete, testable artifacts. This phase primarily concerned Concept A, developing it into a high-fidelity interactive proof-of-concept prototype, where B and C were developed to a conceptual medium-fidelity level. The purpose of the implementation was to simulate realistic interaction flows and interface structures to enable usability evaluation and informed design reflection.

4.6.1 Prototyping Strategy

Implementation followed a progressive prototyping strategy, moving up from low-fidelity wireframes towards a high-fidelity prototype. The low-fidelity prototypes were paper-based wireframes and annotated sketches, which were then translated into clean, static digital mock-ups using Figma. These digital frames functioned as visual representations, allowing layout, hierarchy, colour schemes, and navigation structure to be review in a more consistent format than paper prototypes. Eventually, Concept A was developed into a high-fidelity interactive prototype in Figma. The prototype included clickable navigation, scrollable content, overlays, dropdown menus, and basic interface states. Several components were designed to simulate system

behaviour, such as predefined test results being added when a user selects "add test" action, or status indicators updating as steps in the workflow are completed. Although the prototype does not support true data entry or backend logic, these simulated interactions were found to be sufficient in conveying the intended system behaviour and user flow. This approach aligns with usability prototyping practices, where perceived functionality often is sufficient for early-stage evaluation (Nielsen, 2012).

The concepts, especially A, were developed with a mobile-first approach, due to the intended shop-floor usage context. Screen dimensions were based on handheld devices (~6.1') comparable in size to industrial scanners, which supports quick scanning and short interaction cycles. The iconography and visuals for the mock-ups and interactive prototype – in terms of learnability – were applied from Google Material's (Google Design, 2016) publicly available library.

4.6.2 Concept Selection and Scope of Implementation

Three parallel concepts were developed during the project, A, B, and C. Due to time constraints and the complexity involved in developing interactive prototypes, only concept A was taken forward into a high-fidelity, interactive prototype. Concept A was prioritized as it represents the most relevant and interaction intensive use case, where usability, and data accessibility are most critical for decision-making and DPP logging. Concept B and C were implemented as mock-ups, including defined pages, components, personas, and workflows, but without prototyping. This prioritization enabled deeper exploration of interaction design and usability testing within the available project timeframe.

The implemented prototype is intentionally limited in scope where real data processing, system integration, authentication logic, or backend interoperability is not included. Input fields are mostly static, and system responses are simulated rather than computed. These limitations were partly because Figma is designed to create prototypes rather than fully functional systems and were accepted as necessary trade-offs to focus on interaction design and perceived usability rather than technical feasibility. The prototype should therefore be understood as a proof-of-concept intended to explore user experience and interaction principles.

4.6.3 Design Decisions and Iteration

Design decisions during implementation were primarily made by the author, informed by literature, interviews, workflow mapping, and iterative sketching. Informal discussions with the RemaNet project representative and academic supervisor were done to validate assumptions, reflect on design direction, and clarifying contextual constraints. Iteration was done continuously throughout the implementation phase, where the interactive prototype was repeatedly adjusted based on reflections from earlier sketches, annotations on printed wireframes, informal discussions, and ongoing comparison with the remanufacturing process. Guided by principles of usability and consistency, components were added, removed, adjusted, or simplified to reduce cognitive load and maintain clarity. Iterative refinement is an established design practice, where implementation and evaluation inform each other in short feedback loops rather than linear progression (Wikberg Nilsson et al., 2017).

4.7 USER TESTING AND EVALUATION

Following the development of the interactive prototype for Concept A, a user testing phase was initiated (Figure 5). The purpose of this evaluation was to assess usability, clarity, perceived usefulness, and trustworthiness of the Digital Product Passport

(DPP) platform from a user perspective. The test was also intended to identify potential adoption barriers at an early stage and was exploratory in nature, aimed to support reflection and design recommendations rather than statistical validation or final optimization. Given the time constraints of the project and the thesis scope, the results of the user testing were not used to drive further iterative redesign, instead it informed the discussion and recommendations for future consortium development.



Figure 6. *User Testing.*

4.7.1 Cognitive Walkthrough

The evaluation was conducted as a cognitive walkthrough (introduced by Wharton et al., 1994) combined with usability testing. This approach was selected because it's well suited for early-stage interactive prototypes, where the focus lies on understanding whether users can intuitively navigate the system, understand its feedback, and complete tasks as intended without prior training. A cognitive walkthrough emphasizes step-by-step task completion from the user's perspective, allowing the researcher to observe how users interpret interface cues and system states. The walkthrough was complemented by qualitative usability testing, where verbal feedback, behavioural observations (screen movements, confusions), and post-task reflections were collected to gain insight on the user experience.

4.7.2 Participants and Test Context

A total of six participants took part in the evaluation (see Figure 6). All participants were recruited based on their relevance to the broader remanufacturing ecosystem, consisting of engineers, service, and repairers. To ensure anonymity, participants are referred to as User 1-User 6 in this thesis. All tests were conducted in person, with participants using their own mobile phones to access the Figma prototype via a shared link. Each test session lasted approximately 20 minutes and was moderated and observed by the author. Participants were informed that:

- The system was a prototype,
- The purpose was to evaluate the design (not the user),
- And that incomplete or unclear functionality was expected.

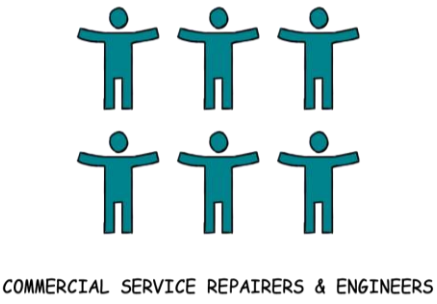


Figure 7. *Participants.*

4.7.3 Test Scenario and Tasks

All participants were given the same scenario and role at the beginning of test, where they were asked to imagine themselves as remanufacturing operators that were going to be working with a faulty product (an electric motor). A task list was used for all participants, structured to follow the linear workflow of Concept A, including:

- Registering an incoming core,
- Reviewing inspection information,
- Navigating disassembly guides and component handling,
- Reviewing testing steps and results.

The full task list can be found in Appendix D together with the remainder of the protocol. Although tasks were predefined the walkthrough was guided, meaning the moderator introduced each task sequentially to maintain consistency across participants, who were encouraged to proceed independently while verbalizing their thoughts when possible (referred to as a *Thinking Aloud* (Nielsen, 2012)). The moderator remained mostly silent during task execution, intervening only to clarify task instructions when necessary or to remind users to explain what they were thinking.

4.7.4 Data Collection

Multiple qualitative data sources were collected during testing sessions to ensure a rich understanding of the user interaction:

- **Direct observation:** hesitation, confusion, misinterpretation, and navigation behaviour.
- **Verbal comments:** spontaneous reactions and reflections during task execution.
- **Post-task questions:** addressing perceived clarity, understanding, confidence, and expectations.
- **A short final questionnaire:** focusing on trust, usefulness, information clarity, and likelihood of remanufacturing industry adoption.

The test sessions were audio recorded with participants' consent to support accurate interpretation and later summarization of feedback. No personal or identifying feedback was stored.

4.8 METHOD DISCUSSION

This section reflects on the methodological choice made throughout the project up to the prototyping phase with focus on credibility, reliability, and limitations of the applied methods. The purpose of this discussion is to critically evaluate how well the selected methods supported the research questions and to clarify methodological trade-offs made during the design process.

4.8.1 Credibility

To increase the credibility of the study, multiple complementary methods were used across all phases of the project. Literature review, interviews, workflow mapping, ideation techniques, and iterative prototyping were all combined to build a coherent understanding, which helped validate insights across theory, empirical input, and design exploration. Challenges related to fragmented product information and inefficient data access, for example, emerged consistently across academic literature, (European Union's) policy documents on the DPP, and stakeholder interviews. These insights were subsequently translated into workflow models and interface requirements (paired with necessary components/elements), strengthening the connection between problem identification and design.

4.8.2 Scope of Empirical Input

A limitation of the context immersion phase is the restricted number of formal expert interviews conducted. While the interview provided in-depth qualitative insight into remanufacturing practices and data challenges, a broader interview sample could have improved the study. This limitation was partially mitigated through further literature review, informal expert discussions with supervisors and project partners, and continuous validation of assumptions during the design iterations. Depth of understanding was prioritized over statistical generalisation, which is consistent with design-driven research approaches – this, however, means that findings should be interpreted as contextually grounded rather than universally applicable.

4.8.3 Prototyping

The implementation phase relied on an interactive prototype developed in Figma. The prototype simulated system behaviour through predefined interactions (based on remanufacturing workflow), visual feedback, progress visualization, and navigation flows, it did however not include functional backend logic or live data handling. This represents a methodological trade-off, where the prototype was intentionally designed to evaluate usability rather than technical feasibility. According to usability practice, interactive prototypes are suitable for identifying usability issues and evaluating user acceptance in early design phases (Nielsen, 2012). Although data input and handling would be preferred to simulate a further realistic scenario while evaluating the prototype, Figma also has its limitations on what can be accomplished in the program. The lack of real system behaviour may nevertheless limit conclusions related to performance and robustness.

4.8.4 Concept Prioritization

Although three concepts were developed parallel to mock-up-level to address different stakeholder perspectives, only Concept A was implemented as an interactive prototype. The prioritization was driven by time constrains and the ambition to explore one concept in depth rather than superficially developing all three. From a methodological point of view, this is a conscious trade-off between breadth and depth. Concept A was chosen due to its alignment with the main research question, RemaNet's focus on creating an attractive DPP-enabled, and being the concept of the stakeholder groups that interact directly with the DPP. Concepts B and C were therefore only developed to mock-up stage.

4.8.5 Iteration

The design process was highly iterative, with frequent movement between wireframing and prototyping. Earlier artefacts were continuously revisited, annotated, revised, and sometimes discarded as understanding of user needs and system constraints evolved. This approach supports methodological robustness by allowing assumptions to be questioned and refined over time, however, the author's dual role as designer and researcher introduces a potential risk of bias. This was addressed by grounding design decisions in theory, documenting design rationales, and seeking external feedback from supervisors and project partners during the iterative process.

4.8.6 User Testing

User testing was conducted as a final validation step of the interactive prototype for Concept A. The testing was a cognitive walkthrough usability evaluation, where six participants with professional backgrounds in repair services and engineering took part of the study. From a methodological standpoint, the user testing strengthens the credibility of the outcomes by providing empirical feedback on the usability and acceptance of the proposed solution. However, further development was not done

after the evaluations and was instead synthesized to recommendations and guidelines for future development. This aligns with the exploratory and design-driven nature of the thesis, where the primary objective is to investigate usability and adoption challenges. The user tests would preferably be done with explicitly remanufacturing operators and technicians, but due to the secrecy of certain workflow processes, many candidates preferred to opt out of the study. Usability experts would also be stronger candidates for the study; however, the participants that conducted the evaluation represent realistic end-user profiles like the ideal end-users, which is considered appropriate for formative usability evaluation in design research contexts (Nielsen, 2012). The testing involved guided tasks, observation, think-aloud, verbal comments, post-task questions, and a final questionnaire. This combination enabled both behavioural observations and reflective feedback on the system.

5 Results

This chapter presents the results of the thesis work, structured according to the phases of the applied Double Diamond design process. The results are derived from empirical data collection, analysis, iterative design exploration, and are presented in a way that reflects how insights progressively informed the development of the proposed Digital Product Passport (DPP) platform.

5.1 RESULTS OF DISCOVER & DEFINE

5.1.1 Interview Findings

To understand current challenges, workflows, needs, and expectations related to product data in remanufacturing, a semi-structured expert interview was conducted with a representative from Wexoe. The participant, who is categorized as a *Remanufacturer*-adjacent stakeholder, has several years of experience in sales and operations for remanufactured components which provided an insight into both customer interactions and internal processes.

The interview revealed that remanufacturing is primarily driven by sustainability regulations and corporate CO_2-reduction goals. However, a major pain point was inconsistent access to reliable product information. The participant emphasized that corporate sustainability reporting requirements now compel companies to quantify their environmental impacts, but relevant lifecycle data are often unavailable or fragmented across suppliers. This lack of transparency complicates both decision-making within remanufacturing and verification of sustainability claims. Furthermore, the participant noted that UX and information design were very important for the success factor for any digital platform, since technicians and customer agents already work under high information loads – meaning that a DPP platform must simplify data complexity. A lightweight and mobile friendly interface that provides immediate feedback which is context-specific was described as ideal. Finally, a recurring concern was interoperability. Since many components/products originate from OEMs in different regions (notably Asia), inconsistent data standard and limited traceability reduce confidence in reported CO_2 metrics.

The key themes identified from the interview, categorized by topics and design implications on the DPP platform, are summarized in Table 5.1.1.1.

Table 5.1.1.1. Overview of themes, key findings, and extracted design implications from stakeholder interview.

Theme	Key Findings	Implications
Sustainability as a primary driver	Remanufacturing is mainly motivated by sustainability regulations and CO_2-reporting obligations, rather than cost savings. Companies must demonstrate measurable emissions reductions and report correct data.	The platform should visualize environmental impact (for instance, CO_2 savings dashboards) to reinforce value and compliance benefits.

Data availability and accuracy	Reliable product data is often missing. It is difficult to quantify emissions without consistent lifecycle data from OEMs.	Prioritize data standardization and traceability features. Ensure access to verified sources and metadata.
Usability and information overload	*"We don't want to bombard anyone with information."* Operations already handle complex data and documentation.	Apply UCD principles. Emphasize clarity, context relevance, and progressive disclosure.
Workflow integration	Technicians need immediate access to repair history, disassembly instructions, specifications, etc., without switching systems.	The platform should conceptually support integration with existing workflows (ERP/PLM) *but actual system integration lies beyond the scope of this thesis – which focuses on demonstrating UX principles and apply change management.* DPP platform should provide quick-scan or data access via QR codes.
Trust and transparency	Sustainability reports must be backed by reliable data; incomplete data undermine confidence in remanufacturing outcomes as well as the DPP system and platform.	Include transparent data provenance and mechanisms showing data completeness or confidence levels (honest data).
OEM dependency and interoperability	Many products originate from OEMs outside the EU which often incompatible or missing data standards.	Ensure interoperability through standardized ontologies and alignment with EU DPP and Ecodesign standards.
Adoption and motivation	Employees may resist new systems if they perceive them as extra work; motivation increases when benefits are visible/tasks are simplified.	Incorporate UX design and change management principles (for instance, instant feedback, visible progress, emissions indicators) to foster engagement.

5.2 USER RESEARCH AND INSIGHTS

5.2.1 User Journey and Design Opportunities

Based on insights gathered from stakeholder interviews and the thematic analysis of pain points, a user journey (see Appendix E) was developed to illustrate how remanufacturing technicians and managers currently work and interact with product data. The journey identifies critical touchpoints where the introduction of the DPP and its corresponding platform could improve efficiency and data transparency. This synthesis provides the foundation for the concept development phase that follows.

The user journey illustrates the end-to-end experience of remanufacturing personnel interacting with product data throughout the process. The journey map highlights several friction points which reveal where the DPP and its corresponding platform could simplify information access and improve decision-making. The journey also shows opportunities to enhance trust and a good UX through real-time feedback, mobile accessibility, data transparency, and context-aware information. These findings directly inform the following concept development phase, where three alternative design directions are explored to address the identified needs and pain points.

5.3 CONCEPTS OUTLINE

Building on the findings of the context immersion, theory, user journey and stakeholder interview, three design concepts were developed to explore alternative directions for a DPP platform that supports remanufacturing workflows. Each concept addresses similar challenges but emphasizes a distinct separate interaction focus and stakeholder perspective – together they aim to identify different paths for improving efficiency and value of remanufacturing through the DPP platform. Each concept represents an interpretation of how DPP data could be used and visualized to support the needs of the stakeholders. By comparing multiple directions rather than pursuing a single solution early, the aim is to better understand the balance between functionality and impact. The following subsections outline the three proposed concepts – each building on the same user journey but targets different user priorities and contexts within the remanufacturing process.

Concept A: Efficiency

Concept A focuses on usability (ease of use) and task efficiency for remanufacturing technicians working in operational environments, where the primary objective is to emphasize data retrieval and input during throughout the user journey. Immediate access to essential data through scanning or quick search functionalities is priority. The concept is tailored to shop-floor conditions where clarity and reliability are critical to quickly assessing cores and to ease the reman process. By streamlining the frequent interactions and reducing cognitive load associated with data entry, the concept supports smoother operations and helps mitigate delays caused by missing or inconsistent information.

Concept B: Impact

Concept B targets managerial and sustainability-oriented stakeholders who oversee remanufacturing performance and circularity metrics. The focus for this concept is for the user to be provided with visualizations of environmental and operational impact through interactive dashboards, where they can monitor progress toward sustainability goals. The emphasis is on making abstract data tangible and actionable, where it's not focused on daily operations like concept A, but more focused on how

DPP data can be visualized and served as strategic evidence for decisions-making and reporting (communication with customers or regulatory bodies).

Concept C: Transparency

Concept C explores a customer-facing DPP-enabled experience focused on transparency, communication, progress tracking, and sustainability feedback. The concept enables end users to follow their product through the remanufacturing process and understand its environment impact without exposing internal operational complexity or "leaking" classified information about the product.

The three concepts represent complementary perspectives – where A optimizes efficiency and usability at the operational level; B focuses on visibility and impact at managerial level; and C on tracking and data transparency. These alternatives form a clear connection to the thesis goals.

5.4 RESULTS OF DEVELOP

This phase focused on detailed design on the selected concepts based on previous insights. Personas followed by a mapped workflow for each concept corresponding to their stakeholders were developed. The foundation of all concepts is established in phase, together with paper-prototype wireframes followed by static mock-ups in Figma. Lastly, the concept that moves forward to further interactive development is decided.

5.5 CONCEPT A

5.5.1 Persona: Remanufacturing Operator

To contextualize Concept A and anchor its design in the reality of remanufacturing – a user persona was developed based on insights gathered from the interview with Wexoe (an authorized Rockwell distributor), CIRPASS (2024), and literature on UX.

Lars Larsson – Remanufacturing Operator:

Table 5.5.1.1. Persona Lars Larsson with his attributes paired with short descriptions.

Attribute	Description
Role	Remanufacturing and repair operator at Rockwell Automation.
Age	38
Experience	12 years in industrial automation and electrical equipment remanufacturing.
Education	Vocational training in industrial maintenance.
Digital Maturity	Moderate: comfortable with mobile tools but avoids complex applications and interfaces.
Primary Tasks	Inspecting returned products, disassembly, diagnosing faulty components, performing part replacement, and testing functionality.
Tools Used	Multi meter, torque tools, grip tools, electrical plier tools, circuit finder, hammers, utility knifes, ERP job

	tracking, manufacturer PDFs/manuals, company PC, tablet, pen, and paper.
Environment	Shop floor, limited desk space, intermittent network connectivity, tablet for documentation.
Goals	Complete remanufacturing jobs quickly and accurate, ensure traceability of each unit, and reduce time spent searching for data.
Pain Points	Missing or outdated manuals, inconsistent data formats, generating reports manually, uncertainty about successful data logging.
Motivations	Lars prides himself through precision and quality when working, he values tools that "just works" and doesn't require extra explanation.
Information Needs from a DPP Platform	Fast product identification (UPI, QR). Verified specifications, tolerances, and safety warnings. Disassembly instructions. Component-level data and test requirements. Confirmation that all steps are logged and traceable
Relevance to Concept A	Lars represents the core user group for which the efficiency-focused DPP platform is designed, who prioritizes speed, clarity, and reliability.

This persona illustrates the operational context in which the efficiency-focused platform must function ensuring that technical functionality supports actual workflow behaviour rather than abstract system goals. The primary user of Concept A is a remanufacturing operator working on the shop floor, exemplified by Lars Larsson. Lars embodies the typical user profile, where field insights indicate that the user operates in an environment with high throughput while often handling several cores in parallel. Lars requires fast and unambiguous access to task-relevant data in the shopfloor where connectivity and available hardware (tablets, PCs, etc.) may vary. Thus, concept A is aimed at being a mobile-first, easy-to-use, low-latency interface that supports quick scanning, local caching, and minimal manual input.

5.5.2 Data and Interoperability Considerations

Although this thesis does not include full system integration, it is architecturally aligned with DPP interoperability standards (CIRPASS, 2024). The data structure assumes use of standardized identifiers and includes fields for:

- Origin, model/ID, manufacturer,
- Material composition, hazardous substances, disassembly instructions,
- Environmental footprint, energy efficiency, recycled content,
- Repair history, usage data, warranties.

5.5.3 User Workflow Summary

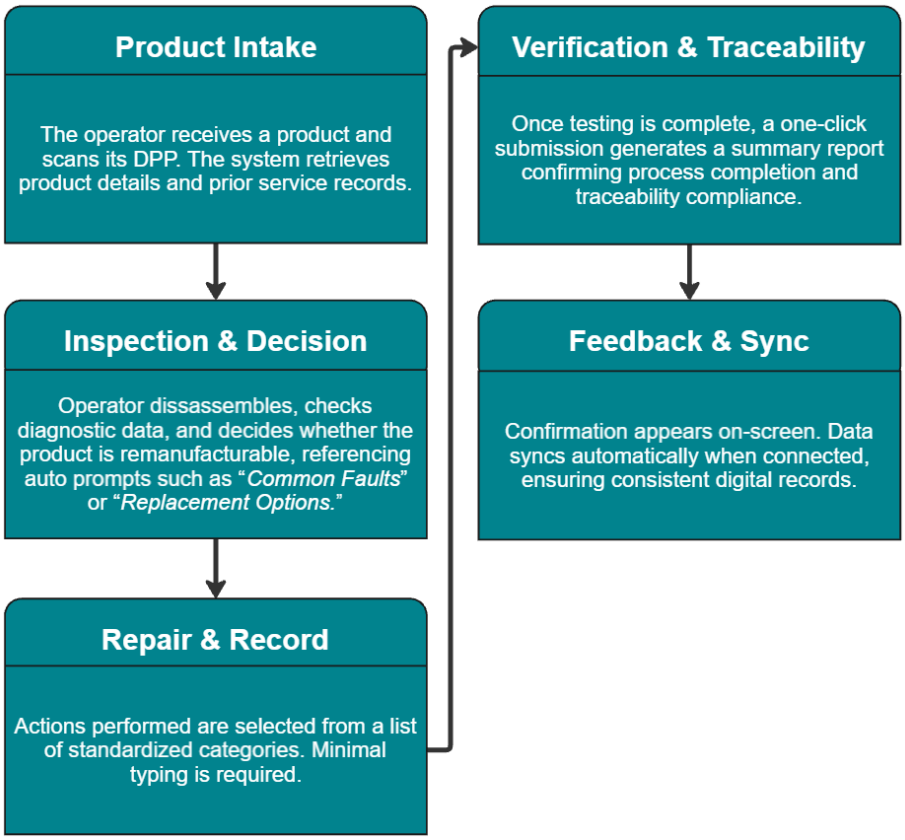


Figure 8. *Concept A user workflow for remanufacturing.*

5.6 CONCEPT B

5.6.1 Persona: The Sustainability & Operations Manager

To ensure that Concept B aligns with the needs of managerial stakeholders, a dedicated persona was developed. This persona synthesizes insights from interviews, literature on managerial needs, and DPP-specific requirements. While Concept A focuses on operational technicians, B targets users with supervisory responsibilities. The persona below represents a typical operations manager who interacts with the DPP system at a strategic level.

Anna Annsson – The Sustainability & Operations Manager:

Table 5.6.1.1, Persona Anna Annsson with her attributes paired with short descriptions.

Attribute	Description
Role	Sustainability and Operations Manager (Remanufacturing Facility).
Age	42
Experience	15+ years in operations and process improvement across manufacturing and remanufacturing sectors.
Education	B.Sc. in Sustainable Production
Digital Maturity	High: daily use of ERP, QMS dashboards, CSR reporting tools,

	SharePoint, Excel, and BI software.
Primary Tasks	Oversee daily remanufacturing operations, ensure production efficiency, track KPIs (throughput, failure rate, CO_2 savings, resource usage), prepare sustainability reports, communicate with OEMs and regulators, identify improvement opportunities in workflows, and data management.
Tools Used	Desktop (primary), laptop, mobile for quick checks. Time available for review: short, fragmented.
Goals	*Operational:* Ensure stable workflow and minimize bottlenecks, detect issues early. *Sustainability:* Report CO_2 savings and environmental impact using reliable data, customer reporting. *Strategic:* Make data-driven decisions and coordinate across teams.
Pain Points	Inconsistent or missing OEM data makes sustainability reporting difficult. Manual data gathering is time consuming and error prone. Lack of cross-OEM standardization makes comparing impact hard. Difficulty verifying accuracy of CO_2 or lifecycle data. Technicians often skip documentation due to workload.
Information Needs from a DPP platform	High-level operational overview (delays, etc). Accurate, aggregated sustainability metrics (CO_2 saved per product, per year). Traceability data such as component history, reman events, testing outcomes. Alerts on missing data, incomplete processes, or late tasks.
Motivations	Provide accurate CSRD-compliant sustainability reporting. Improve workflow efficiency using data. Reduce costly errors and rework. Strengthen the facility's competitiveness with data-driven insights.
Frustrations	Systems that require too much navigation or *"too many clicks"*. Raw technical data with no aggregated insights. Platforms without cross-OEM interoperability. Slow or unresponsive dashboards.
Task Context	Works from office, sometimes on the shop floor.

This persona serves as the foundation for the structure of Concept B. Goals, pain points, and information requirements will guide the selection of components and

visualizations for the concept, ensuring that the platform supports managerial responsibilities. Concept B is designed for remanufacturing managers who require an overview-level interface rather than task-level guidance. Unlike operators, these users do not work directly on the shop floor but instead they rely on aggregated data to make strategic decisions. The typical user, represented by persona Anna Annsson, operates in a context where she manages multiple ongoing jobs, monitors throughput, evaluates CO_2 savings, and communicates results internally and externally. Her work is characterized by:

- Fragmented data spread across ERP systems, Excel sheets, PDFs, and supplier portals.
- Increasing pressure from EU sustainability regulations.
- A need to justify remanufacturing value to customers and upper management.
- Responsibility for ensuring standard compliance, documentation, tracking, and traceability.

Concept B addresses this context by providing a dashboard-driven interface with drill-down capabilities for more detailed views. The interface emphasizes clarity and comparability, through automated data visualization. Because of the differing digital literacy levels, and higher cognitive load from decision-making responsibilities, the design prioritizes readability, reduction of noise, adjustable detail levels, and data provenance.

5.6.2 Data and Interoperability Considerations

Concept B handles data at the aggregated and cross-product level, rather than at individual component resolution. Its structure aligns with DPP interoperability requirements described by CIRPASS (2024) and upcoming ESPR mandates but is adapted for managerial support.

5.6.3 User Workflow Summary

While operators follow a sequential workflow through the remanufacturing process, managers operate in a non-linear monitoring and decision-making loop, structured as follows:

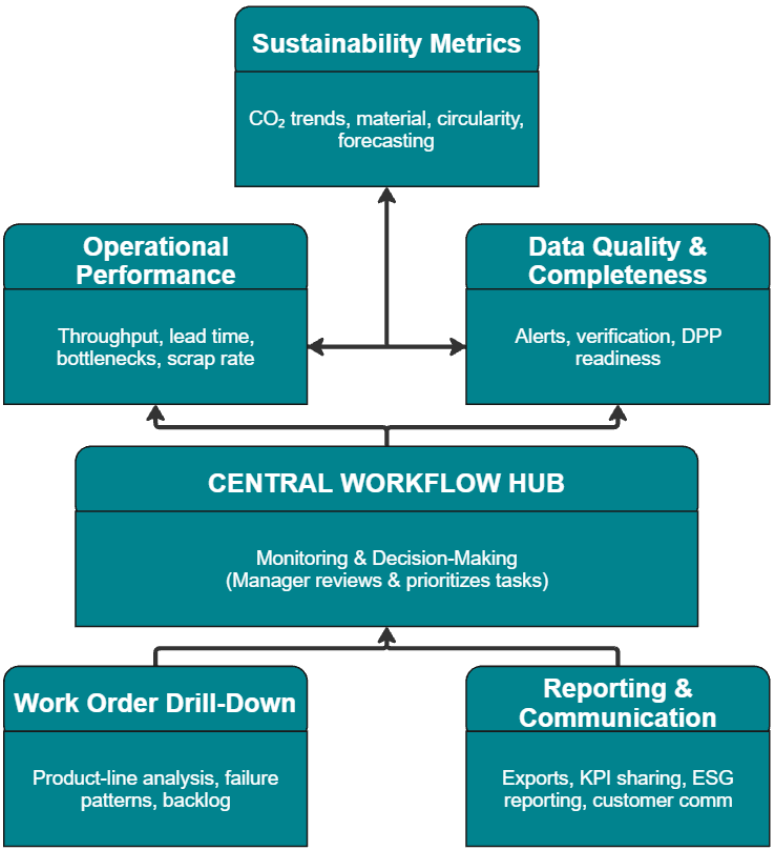


Figure 9. *Hub-and-spoke diagram of Concept B users' non-linear workflow.*

5.7 CONCEPT C

5.7.1 Persona: Maintenance Manager (Industrial End-User)

The primary user for Concept C is an external industrial customer, represented by persona Ali Alsson (below). Unlike Concept B and C, which are focused on remanufacturing operators and managers, C addresses end-users who own the product and initiate remanufacturing requests, who depend on reliable information during machine downtime.

Ali Alsson – Maintenance Manager:

Table 5.7.1.1, Persona Ali Alsson with his attributes paired with short descriptions.

Attribute	Description
Role	Maintenance Manager at a mid-sized manufacturing company.
Age	44
Experience	18 years in maintenance operations, including procurement and coordination of external service providers.
Education	B.Sc. Mechanical Engineering + industry certificates in maintenance planning
Digital Maturity	High: uses ERP and CMMS daily.
Primary Tasks	Overseeing machine uptime, planning

	maintenance cycles, coordinating repairs/returns, communicating with suppliers and remanufacturers.
Tools Used	ERP (SAP), CMMS (maintenance management software), mobile phone for tracking deliveries, email, spreadsheets.
Goals	Ensure machines are restored quickly, track the remanufacturing status of returned equipment, document CO_2 savings for sustainability reporting, validate warranty, and service quality.
Pain Points	Limited visibility into remanufacturing progress, unclear return timelines, difficult communication with remanufacturers, incomplete sustainability documentation, pressure from management to prove reductions in environmental impact.
Motivations	Fast turnaround, transparency, predictable schedules, sustainability KPIs, clear communication.
Task Context	Office-based but frequently on the production floor; responsible for minimizing downtime during failures.
Data Access Level	Restricted user: can see product-level status, carbon savings, warranty, and service outcomes, but not internal remanufacturing measurements or component-level diagnostics.
Relevance	Represents the typical customer-side user who initiates the remanufacturing request and needs high-level visibility. Track progress, ensure transparency, and receive sustainability documentation.

Ali Alsson operates in a fast-paced industrial environment where equipment failures directly impact production efficiency. His responsibilities include coordinating returns and communicating expected delivery timelines to upstream managers and production teams if maintenance windows occur. His interaction with the *Digital Product Passport* (DPP) centers around high-level visibility and not operational detail.

5.7.2 User Workflow Summary

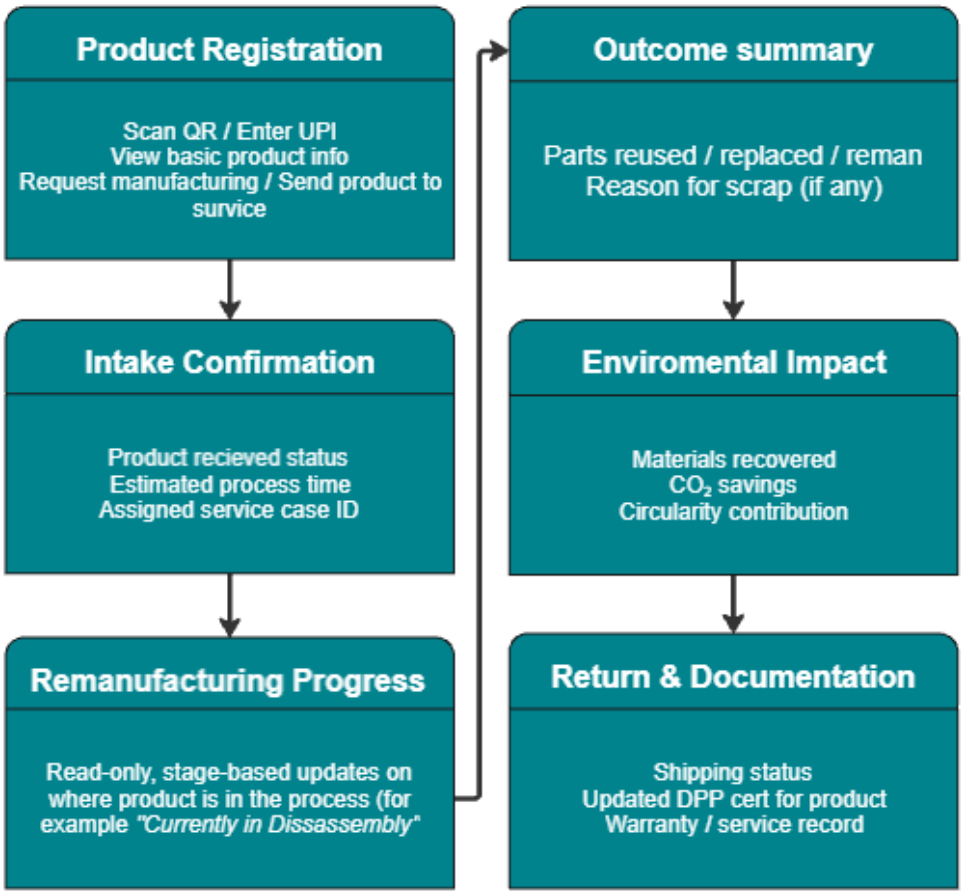


Figure 10. *Concept C user workflow.*

5.8 CONCEPT SYNTHESIS

To translate the user journey and workflow analysis into an actionable interface structure each stage of the process was mapped to corresponding interface components. The following tables, which can be seen on Appendix F, outline the proposed elements for the wireframe of Concept A, B, and C. Through stakeholder interviews, workflow- (Figure 7, 8, and 9) and user journey-mapping, recurring challenges related to data fragmentation, usability, and trust in sustainability reporting were identified. These findings provided a clear foundation for ideation and concept development.

The three design concepts developed in this phase represent alternative interpretations of how a Digital Product Passport platform could support remanufacturing from different stakeholder perspectives. Concept A prioritizes operational efficiency and usability for shop-floor technicians; Concept B emphasizes sustainability impact and performance visibility for managerial stakeholders; and Concept C explores transparency and status communication toward end users. Together, these concepts illustrate how the same underlying DPP data can be structured and communicated differently depending on user role and context.

Concept A was selected for further development. This decision was motivated by its close alignment with the core research questions, which focus on usability, adoption, and interaction at the point where the DPP is actively used. While Concepts B and

C remain valuable as comparative and complementary directions, Concept A offered the most suitable scope for detailed prototyping within the constraints of the project timeframe. Concept B and C did however proceed to a static mock-up level to conclude their development.

5.9 FURTHER DEVELOPMENT

During Phase 3, the Concepts moved on to further development where the established components for each page were translated into paper-prototypes, and subsequently into mock-ups. The end of phase 3 focuses on translating Concept A from a conceptual structure into a tangible, interactive proof-of-concept prototype. The objective was not to develop a technically complete system, but to explore how the identified design principles, workflows, and data requirements could be embodied in an interface that supports remanufacturing work in practice.

5.9.1 Wireframing to Mock-up

Following the definition of interface components and workflow mapping in Phase 3, Concept A, B and C was further developed through an iterative prototyping process. The initial paper-based wireframes (generated through the braindrawing) were used to explore layout, hierarchy, sizes, and information flow across the different pages, or stages, across the remanufacturing process. These sketches (see Figure 10) allowed for rapid iteration, annotation, and refinement of ideas, supporting early decision-making regarding navigation structure and component placement.

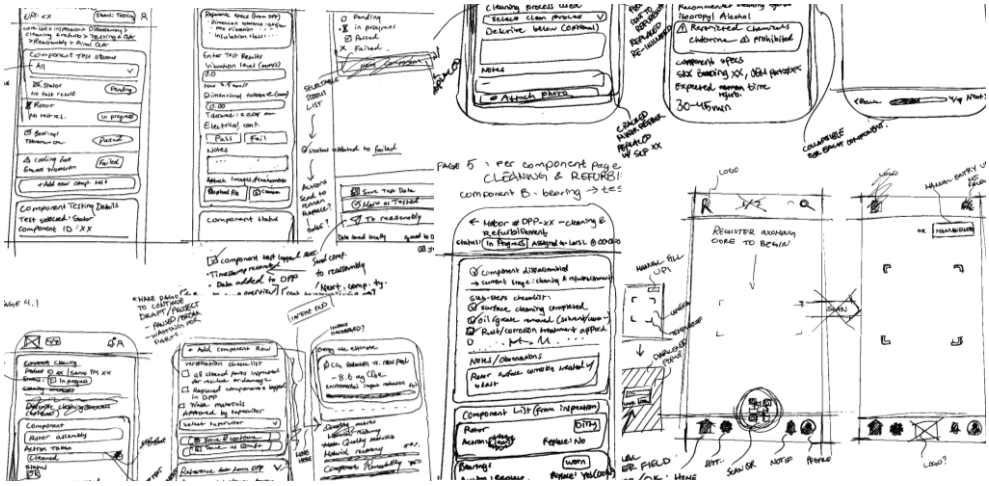


Figure 11. *Examples of paper-prototype wireframes of the concepts.*

The paper wireframes were then translated into medium-fidelity digital mock-ups using Figma, this is where the development of Concept B and C concluded. Their mock-ups nonetheless serve as supporting material, illustrating alternate design directions and informing later discussion. To create the mock-ups, the hand-drawn wireframes were converted into clean, static visual representations placed within device-sized frames. While these wireframes did not support interaction, they provided a clearer overview of spatial structure and alignment with mobile-first constraints. This transition helped reveal layout issues and redundancies that were less apparent in the paper sketches.

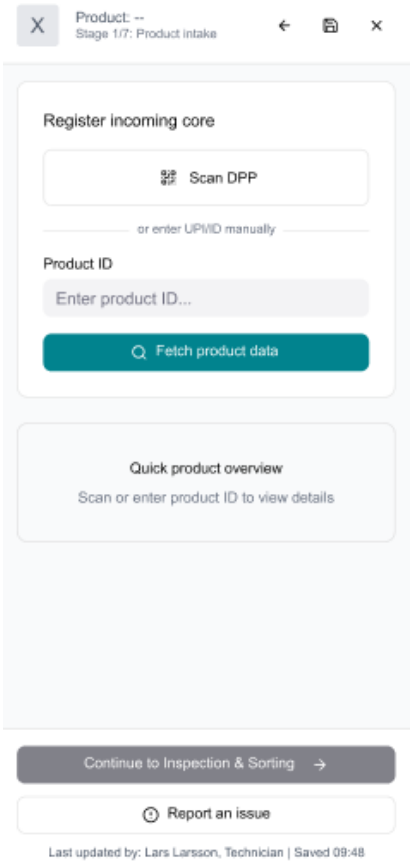


Figure 12. *First page of the Concept A mock-up.*

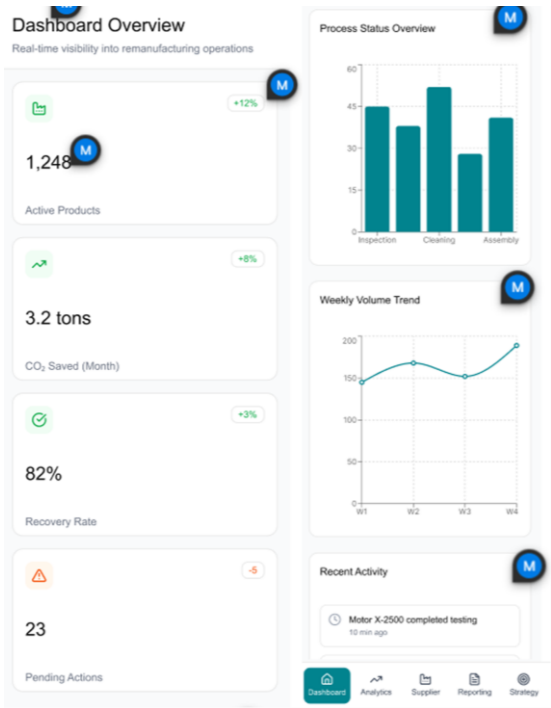


Figure 13. *First page of the Concept B mock-up (split in two).*

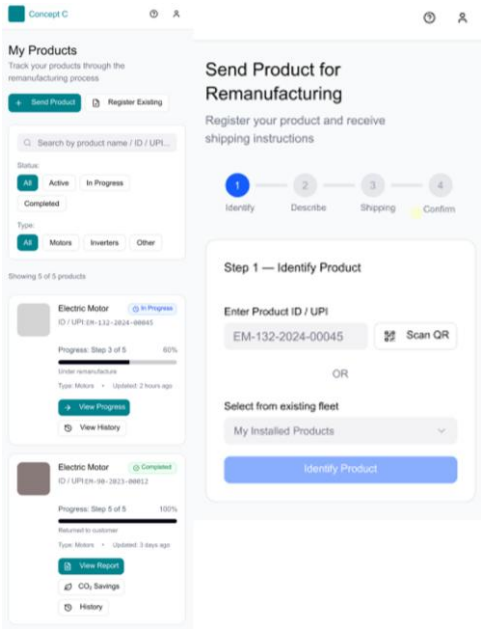


Figure 14. *First pages of the Concept C mock-up.*

5.9.2 Concept A Further Development

Building on these mock-ups, an interactive high-fidelity prototype was developed in Figma for Concept A – where clickable navigation, scrollable content areas, overlays, dropdown menus, and basic state changes were introduced. Several interface elements were designed to "mock" system behaviour, for example pressing an "Add test" button (see Figure 14 (right)), the page would insert a pre-filled test result or a new test form, simulating the outcome of data entry without requiring actual backend functionality (see Figure 14 (left)). By simulating these interactions, the prototype was able to communicate system logic and workflow progression despite technical limitations of the prototyping tool. Visual styling with RemaNet's colour schemes was introduced at this stage to support clarity and hierarchy while avoiding aesthetic detail, like other ERP systems, that could distract from interaction and usability.

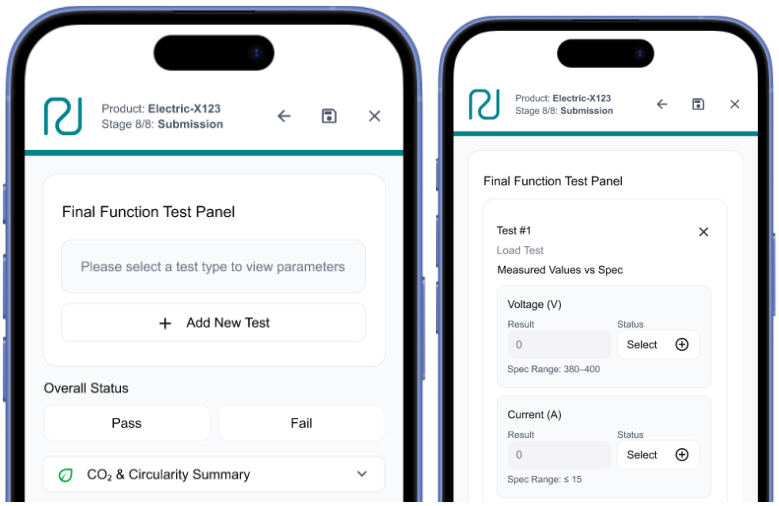


Figure 15. *Add New Test function in Concept A.*

The function of enabling different states, allowing for checking boxes and selecting between elements done by using the new Boolean variables in Figma, together with Component Sets to group components in different states (see Figure 15). This also allowed for visually enabling/disabling a button to continue, for example, depending on whether a "completed" button had been checked or not – this was done through Figma's new If-logic within the Frames and the Component Sets. Other functions, such as the variation of a dropdown (an overlay) for selecting *condition of a component,* utilized both Property-states (like Boolean states) within the component set, which opened an overlay when clicking on the element, allowing the user to select the proper condition of the component. After selecting the condition, the overlay utilizes the function Set Variable Mode to change the component status and then closes the overlay. The logic can be seen below on Figure 16).

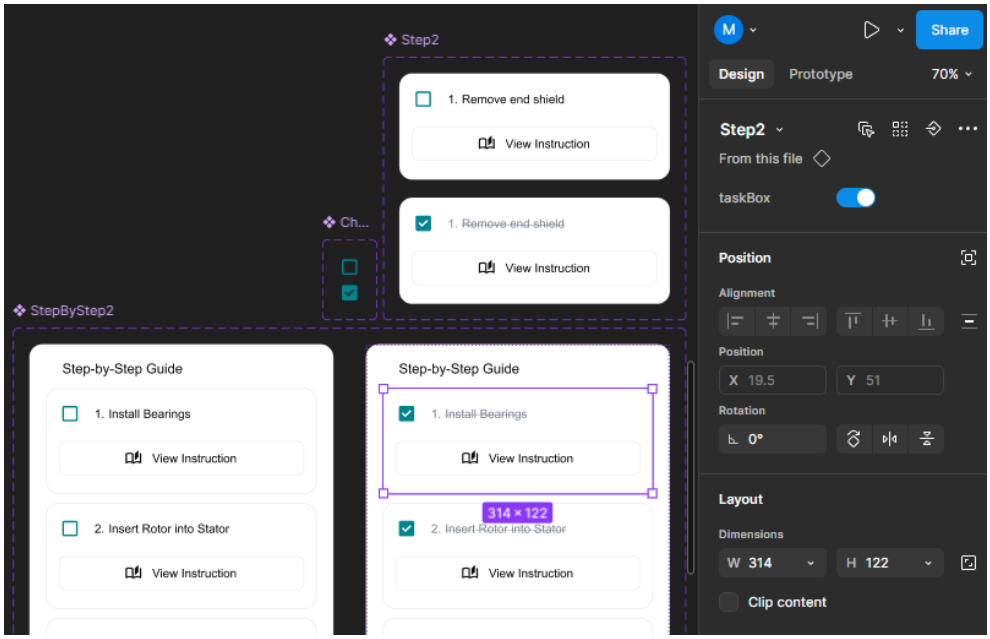


Figure 16. *Boolean states and Component Sets.*

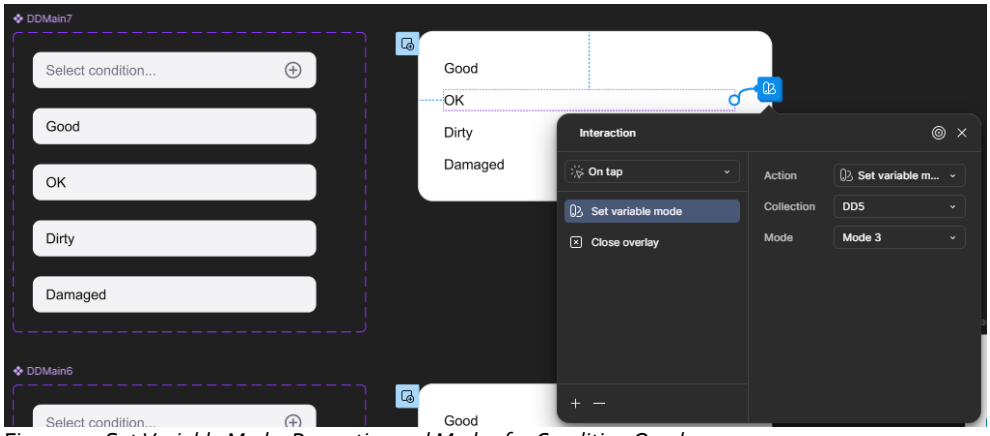


Figure 17. *Set Variable Mode, Properties and Modes for Condition Overlay.*

5.9.3 Iterative Refinement of Concept A

The prototyping process was iterative, following the Double Diamond process. Throughout Phase 3 and 4, earlier artifacts (wireframes, mock-ups), were

continuously revisited and revised in parallel with the interactive prototype. Elements were added, removed, changed, or simplified as usability concerns emerged during intermetal walkthroughs and informal reviews with the supervisor and representatives connected to the RemaNet project. Design decisions were guided by the principles identified earlier in the thesis, including minimizing cognitive load and providing immediate feedback. The focus remained on enabling technicians to progress through the remanufacturing workflow with minimal friction, while still ensuring data points required for traceability and sustainability were consistently logged. The prototype after undergoing several iterations can be seen its project with each respective frame, component, and prototype connections in Figure 17, below.

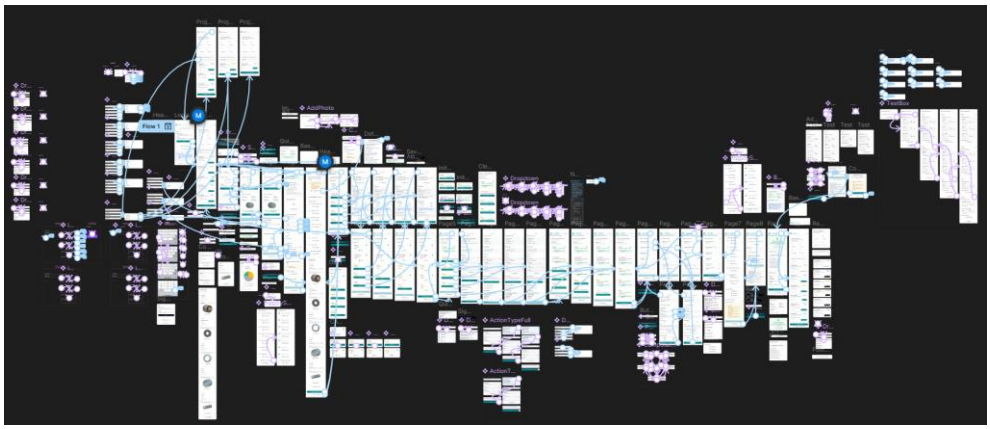


Figure 18. *The Concept A Project in Figma.*

The following frames (Figure 18-36) will consist of the final proof-of-concept prototype pages:

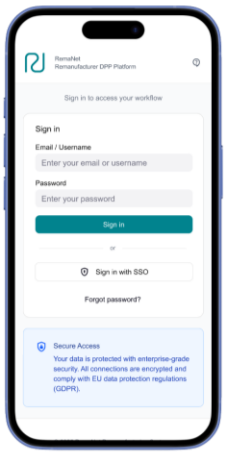


Figure 19. *Login Screen.*

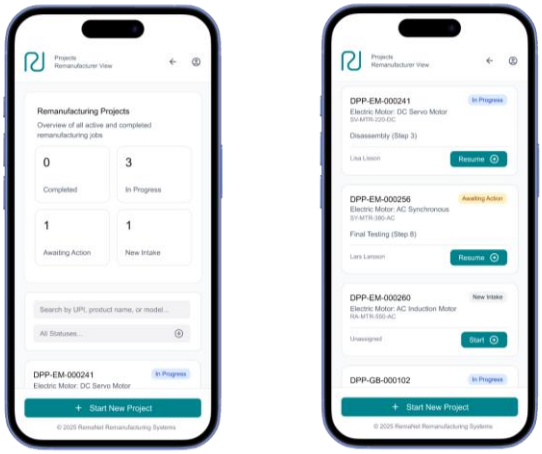


Figure 20. *Dashboard/Home.*

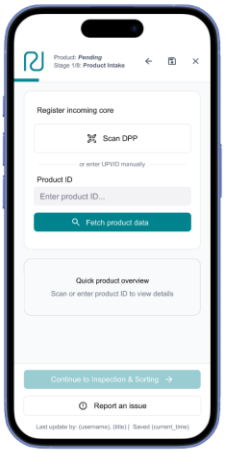


Figure 21. *Product Intake (Stage 1).*

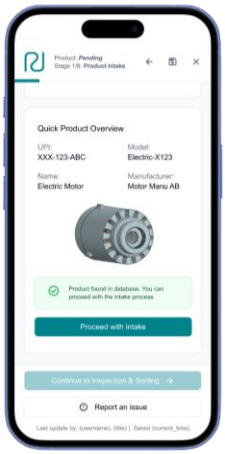


Figure 22. *Product DPP Scanned and Detected.*

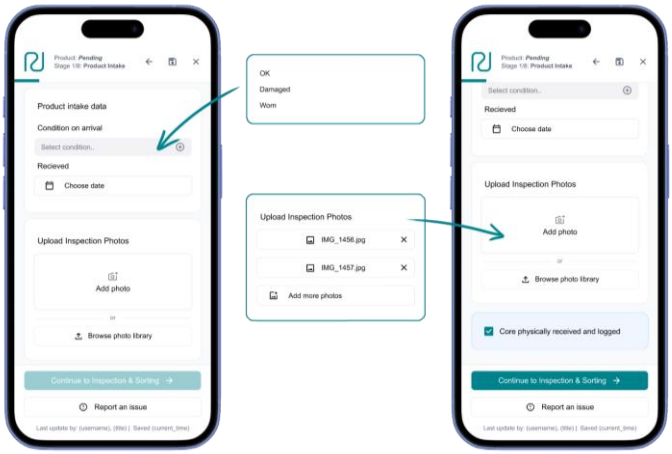


Figure 23. *Registering Product, Selecting Condition, Adding Optional Pictures, and Logging Verification.*

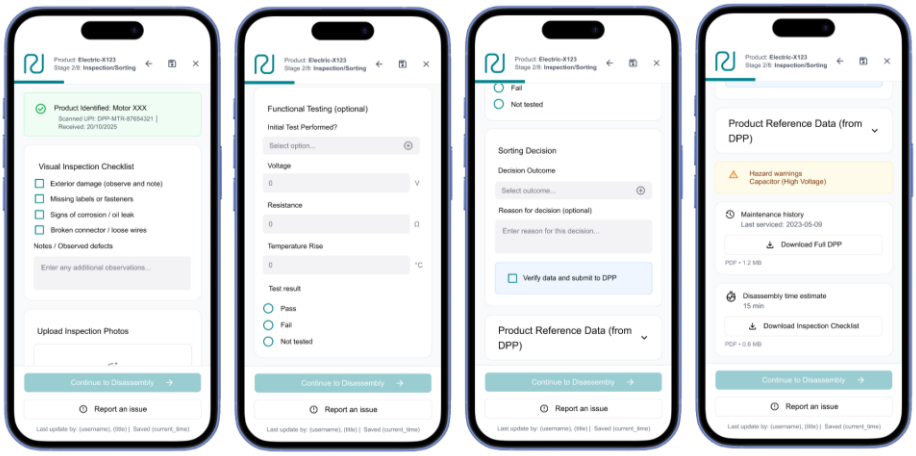


Figure 24. *Inspecting Core.*

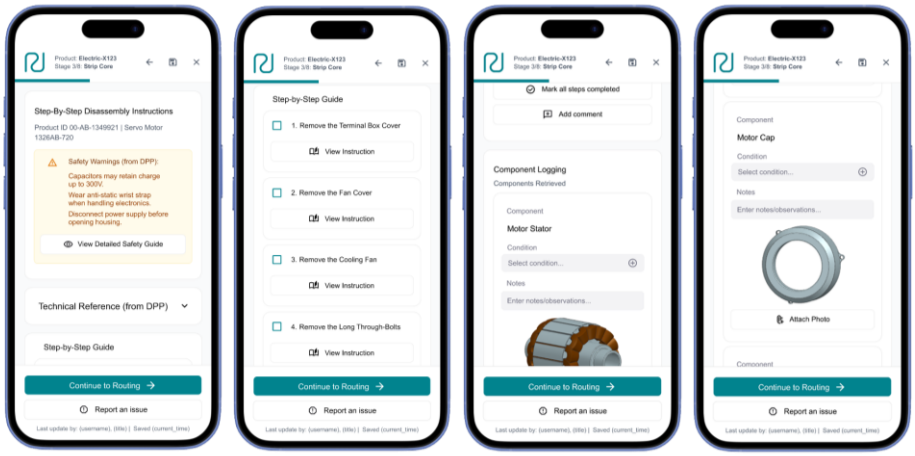


Figure 25. *Stripping Core and Logging Component Conditions.*

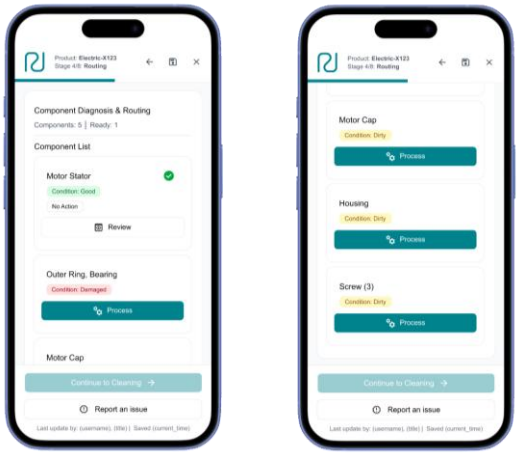


Figure 26. *Deciding Route for each Component.*

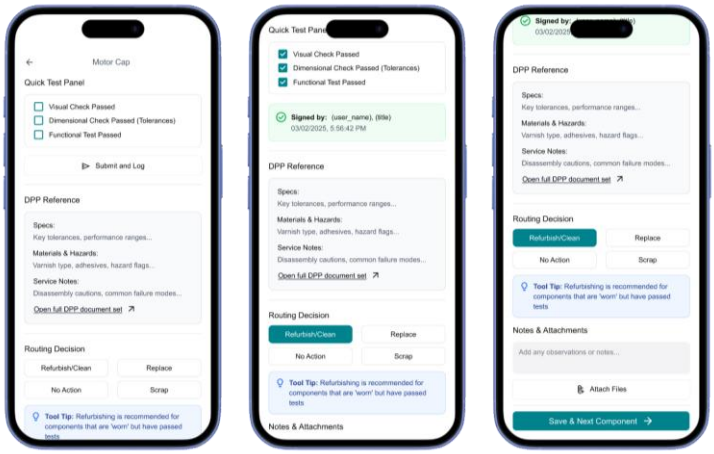


Figure 27. *Component Initial Testing and Routing Decision.*

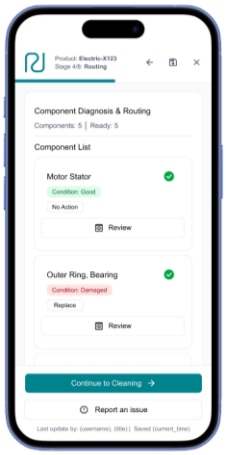


Figure 28. *Routing Page Post-Routing.*

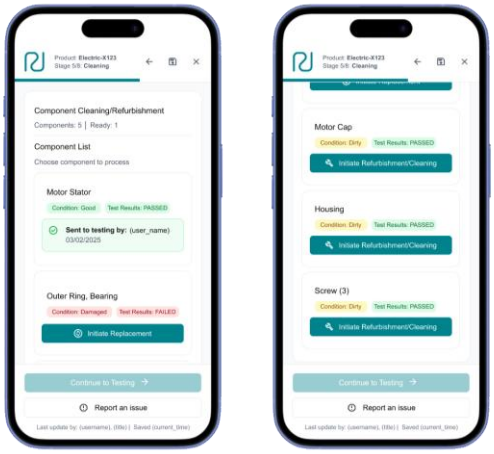


Figure 29. *Component Processing Depending on Routing Decisions (Main Page).*

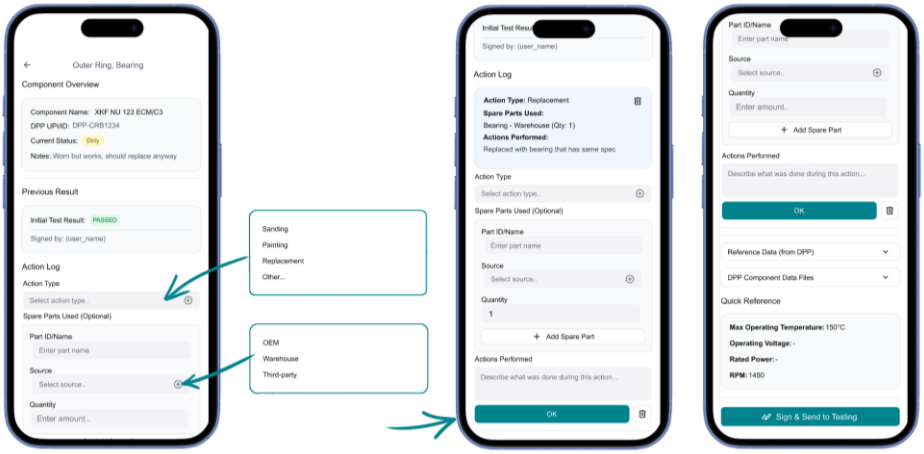


Figure 30. *Individual Component Processing.*

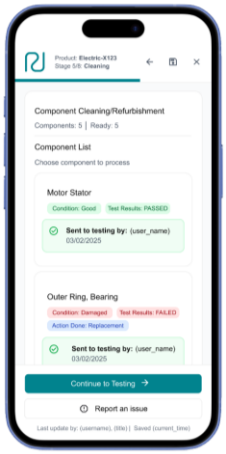


Figure 31. *Processing Done.*

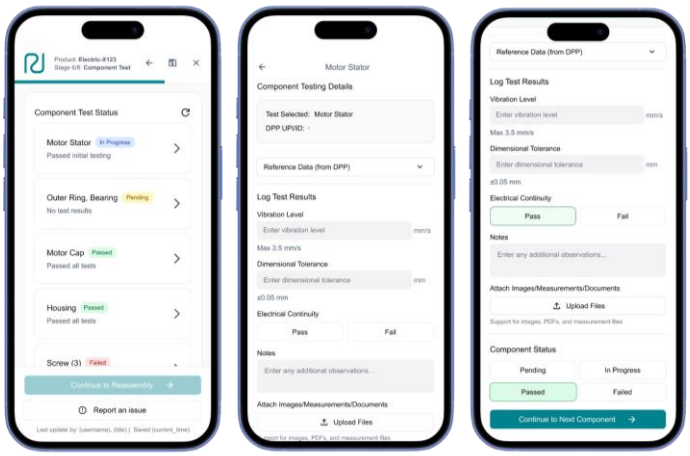


Figure 32. *Component Testing Post-Processing.*

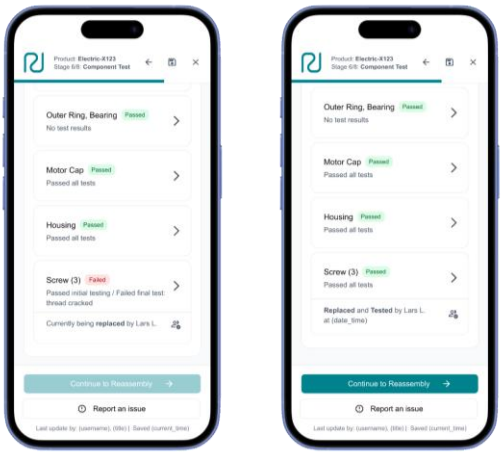


Figure 33. *Component Page Post-Testing. Continue When All Components Tested.*

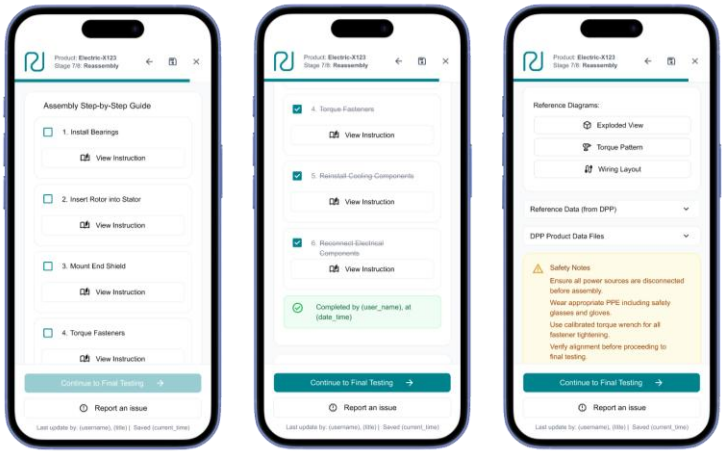


Figure 34. *Reassembly.*

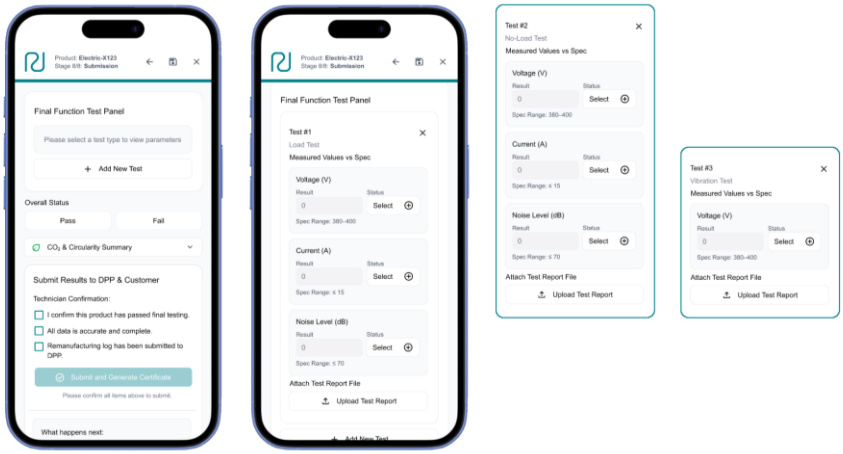


Figure 35. *Final Testing of Assembled Core.*

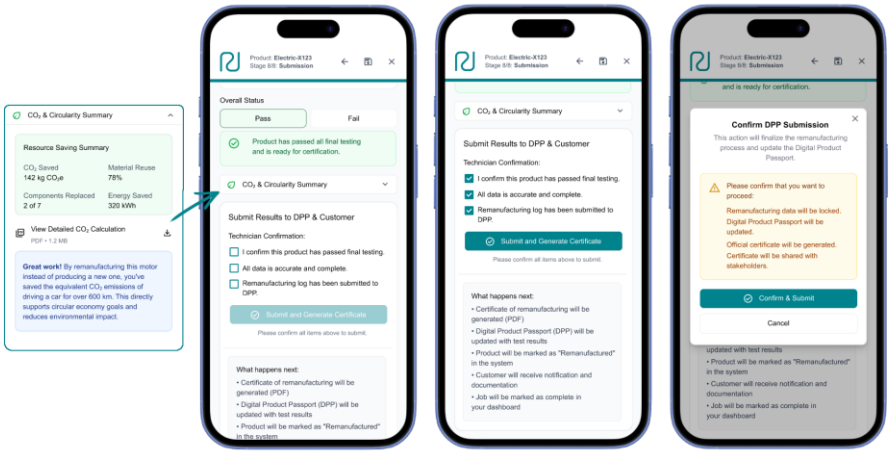


Figure 36. *Submitting Final Testing and Finishing Remanufacturing Process.*

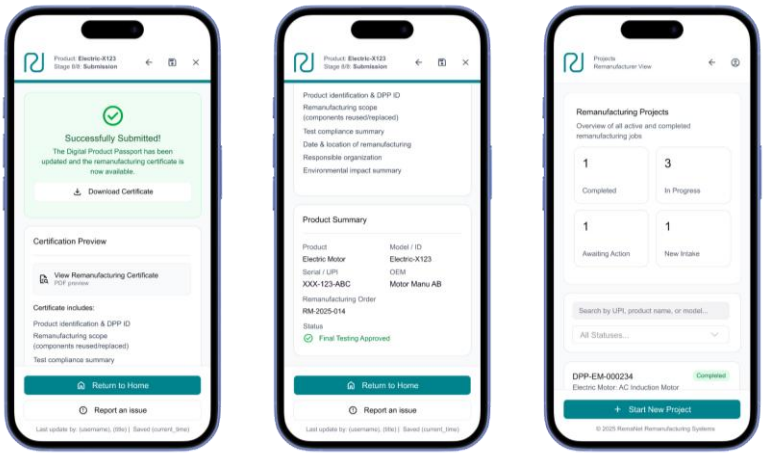


Figure 37. *Submission Receipt and Returning to Dashboard/Home.*

5.9.4 User Journey with Concept A

The user journey extracted together with the remanufacturer workflow for Concept A (see Appendix E), presenting the steps in the remanufacturing process, paired with *Pain Points* and *Opportunities for the DPP Platform*, can now be seen with an added layer: *Platform Solution*, which goes together with the *Opportunities for DPP*

Platform, and Actions layer (see Figure 37).

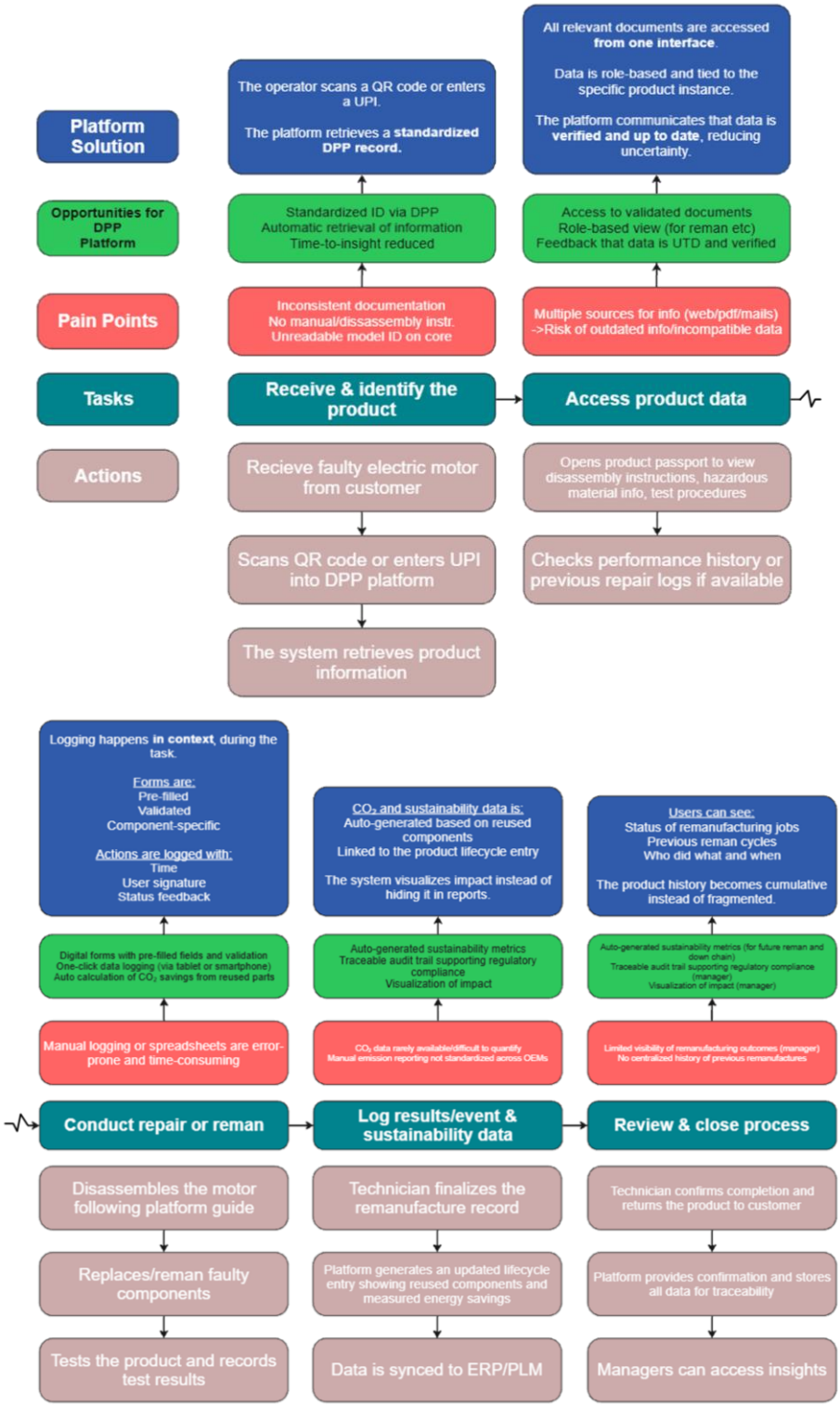


Figure 38. *Updated Reman User Journey with Platform Solution Layer.*

5.10 RESULTS OF USER TESTING

User testing was conducted with six participants (User 1-6) representing a mix of engineers and repair service personnel. All participants interactive with the same interactive prototype of Concept A using their own mobile devices. The test followed a cognitive walkthrough approach with guided tasks and post-task questions. The results are presented below.

Overall Impression

Across all six participants, the prototype was perceived as professional, simple, trustworthy, and suitable for industrial context. Users generally understood the purpose of the system and its role in supporting remanufacturing workflows through structured data input and access to product information. The prototype was described as conceptually strong with clear potential to support daily work if implemented in a real system.

Navigation and Workflow Clarity

A recurring observation across all participants was initial uncertainties while doing the first walkthrough regarding where they were in the process, and when a step was considered completed. While users generally understood the linear structure of the workflow, they sometimes hesitated before advancing, particularly when unsure whether all required actions had been fulfilled, and whether they had to download additional product-related files (PDFs, STEPs) that can be seen on some pages. This issue appeared consistently during participants' first interaction with the prototype but decreased as users became familiar with the flow and layout. Several participants expressed a need for *clearer visual confirmation of task completion as well as more explicit cues indicating the next step in the process.* Some participants got stuck on a particular page where it was required to "refresh" the page to move on, because the user was "waiting" on another operator to finish a task on their project, *unnatural breaks* in the flow like this must be limited. Despite this, all participants reported that the navigation itself was logical and that they were generally able to identify where and how to proceed.

Information and Visualization

Participants perceived the amount of information presented as overall well-balanced, though strongly dependent on user role and technical background. Most participants, who were technically experienced users, found the information level appropriate and aligned with their expectations of working with manufacturing overall. Less technically specialized participants noted that while the terminology was highly technical, it was considered reasonable given the context and target group/domain. No participant described the interface as overwhelming, but some expressed a desire for supporting text or contextual references when interacting with unfamiliar terms.

Trust and Reliability

All six participants stated that the system felt reliable and trustworthy in the way the layout is structured, the visual language, and the step-by-step workflow. No participant reported elements that reduced trust in the prototype. Several users explicitly mentioned that the professional appearance and clarity of the interface increased their confidence in the correctness of the "mock" data being handled, even though they did not personally verify the technical content.

Usefulness and Adoption Potential

Participants consistently expressed that the prototype could support their work in a DPP-enabled workplace by reducing and grouping manual documentation and clarifying component status throughout the remanufacturing process. Users highlighted the value of having structured choices and predefined steps, reducing the risk of missing critical points in the flow. When asked to rate how likely they would be to use such a system regularly, five rated their likelihood as 4 out of 5, while one participant rated it 5 out of 5. The most mentioned barrier to adoption was the learning curve associated with introducing a new system, rather than any fundamental issues with the interface itself.

Suggested Improvements

While no major usability failures were identified, participants suggested several areas for improvement. The suggestions were consistent across participants and were primarily related to improving clarity rather than altering the core structure or functionality of the concept. The most frequently mentioned were:

- Clearer visual feedback when a task or step has been completed,
- Stronger indicators of progress and what remains to be done,
- More explicit visual states or feedback to guide the user forward.

Overall, the user testing served to validate the conceptual direction and interaction principles of Concept A. The findings indicate that the prototype is perceived as useful across different user roles, is understandable, and trustworthy. Due to project time constraints, the test results were not used to further iterate the prototype but instead serve as input for reflection and recommendations discussed in later chapters.

5.11 RESULTS OF DELIVER

The outcome of this thesis is a user-centered proof-of-concept for a Digital Product Passport (DPP) platform intended to efficiently support remanufacturing operators' workflows. The result demonstrates how DPP data can be structured, visualized, and interacted with as an interactive, mobile-first prototype with a primary focus on operational use in remanufacturing environments. The prototype serves as a conceptual exploration of how a future DPP-enabled platform could function from the user experience perspective. The emphasis is placed on clarity and transparency, which are key factors identified through literature review and context immersion.

The prototype is structured around the core stages of the remanufacturing process, with each stage represented as a dedicated interface view. This task-oriented structure ensures that data access and data input are directly aligned with the user's current activity, reducing cognitive load and minimizing unnecessary navigation. In response to user restricted access of information, the user is firstly presented with a log-in screen to verify their identity, where they are later taken to a home page consisting of all current projects and allows the user to create a new one. Allowing users to see all current remanufacturing projects coupled with their individual stages in the workflow allows users to continue others work, or their own work the next day (see Figure 38).

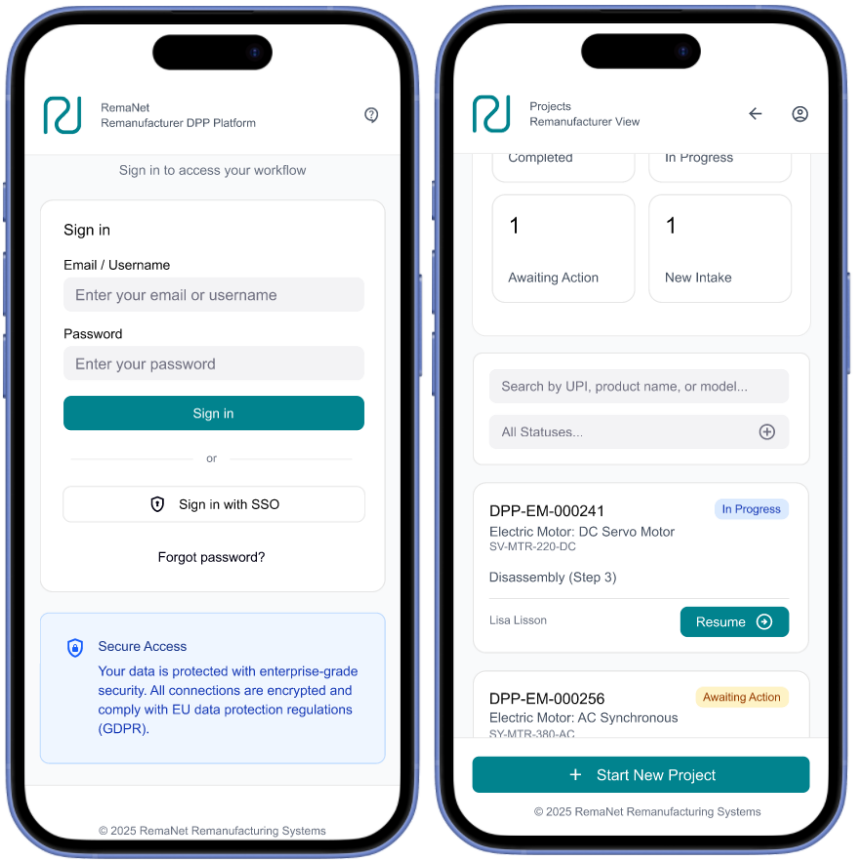


Figure 39. *Login Page (left), Dashboard/Home Page (right).*

Key characteristics of the final concept include:

- Immediate access to product and component data through scanning DPPs or inputting UPIs.
- Structured data input to support traceability and future reporting.
- Real-time system feedback to reinforce user trust and perceived reliability.
- Mobile-first interaction adapted to shop-floor constraints.

The visual and interaction design intentionally avoids excessive complexity, by progressively disclosing information, and only emphasizing task-relevant data at each stage of the workflow. This approach reflects both usability theory and empirical findings, highlighting the importance of simplicity and predictability in high-throughput industrial contexts.

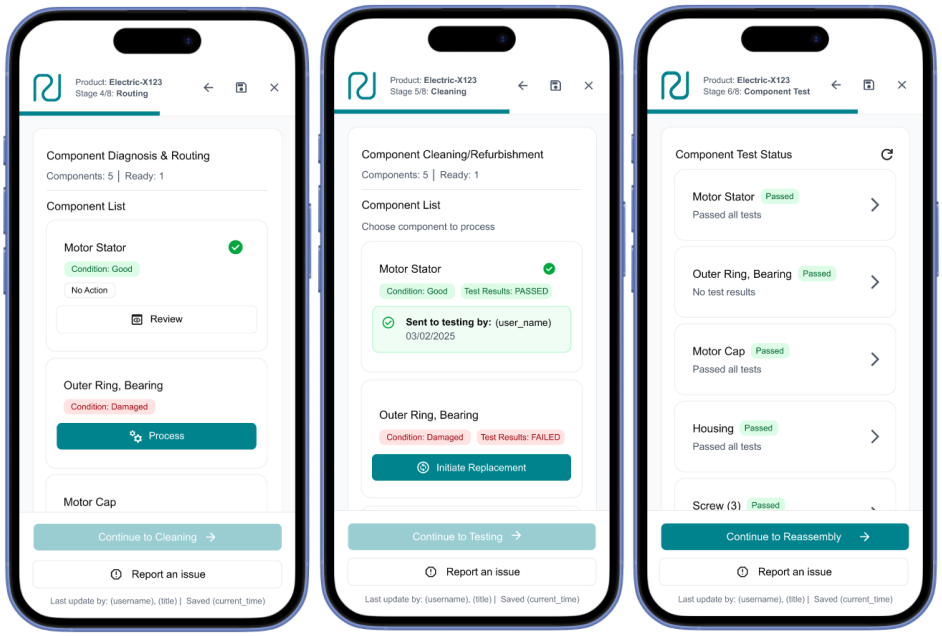


Figure 40. *Routing Page (left), Cleaning Page (middle), Component Test Page (right).*

The final prototype was subjected to formative user testing, which did not result in further design iterations due to time constraints, however, did provide validation of the overall interaction logic, workflow clarity, and perceived usefulness of the concept. User feedback is therefore treated as confirmatory input and as a foundation for the recommendations and future development (covered in the discussion part of this thesis), rather than as a basis for redesign within this project. In addition to the finalized prototype for Concept A – Concept B and C were developed as sketched interface proposals. These concepts illustrate how the same DPP data could be adapted for managerial oversight and end-user transparency. While the concepts were not implemented as interactive prototypes, they complement the (final) result by demonstrating the flexibility of a role-adaptive DPP platform and by broadening the design space explored in the thesis.

Taken together, the result represents a consolidated design proposal which is grounded in theory, empirical insights, and iterative prototyping. The prototype illustrates how a DPP platform can be shaped to support real-world remanufacturing workflows while addressing usability and adoption challenges. The following chapter builds on this result by evaluating the prototype through user testing and discussing implications and opportunities for future work.

6 Discussion

This chapter critically interprets the results of the thesis in relation to its research questions and objectives together with the theoretical framework. Rather than restating findings, the discussion chapter situates the developed proof-of-concept and user testing outcomes within established theories of user-centered design and change management. The chapter also reflects on methodological choices and trade-offs and considers broader implications of designing Digital Product Passport (DPP) platforms through a reflective and analytical perspective. The discussion aims to clarify what knowledge this thesis contributes and how it can inform future DPP development.

6.1 POSITIONING THE RESULT

The primary contribution of this thesis lies in demonstrating how user-centered design principles can be operationalized to make complex DPP-related lifecycle data usable within remanufacturing workflows. The interactive prototype developed in this thesis reflects several core principles from the theoretical framework. The strong emphasis on workflow alignment corresponds to principles of natural mapping and compatibility with user's mental models (Norman, 2013; Preece, Rogers, & Sharp, 2015). By structuring the interface of Concept A around the established remanufacturing process, the prototype reduces the cognitive effort required to translate digital actions into physical tasks and vice versa. This directly addresses one of the central adoption barriers identified in both the literature and the user studies: the perception that new systems, such as the DPP, will introduce unnecessary administrative burden.

The use of progressive disclosure and task-specific views reflects Cognitive Load Theory (Sweller, 1988), which emphasizes the importance of limiting information exposure to what is immediately relevant – it is up to the user in Concept A to expand dropdowns to access further product lifecycle information. User testing confirmed that participants valued the absence of excessive information and appreciated that details were revealed incrementally. This supports Nielsen's (1994) usability heuristics regarding minimalist design and recognition rather than recall, particularly in industrial environments with high tempo. The prototype's feedback mechanisms such as confirmations and progress indicators can be interpreted through interaction design and emotional design theory. Immediate feedback supports visibility of system status (Nielsen, 1994) and contributes to behavioural confidence during task execution (Norman, 2004). Users expressed the need for clearer progress indicators, which further reinforces the importance of state visibility in complex, multi-step workflows, as discussed by Janlert and Stolterman (2010).

From a change management and technology adoption perspective, the results align strongly with Rogers' (1983) Diffusion of Innovations theory. Participants' willingness to use the system was primarily influenced by perceived relative advantages, for example through compatibility with existing workflows, and low perceived complexity. The prototype's scan-first, mobile-oriented interaction model supports these attributes by reducing manual input and aligning with established practices. In this sense, the prototype functions as an illustration of how design decisions can directly support adoption mechanisms described in theory.

Overall, the results provide a concrete response to the main research question by showing that a DPP platform should be structured around user workflows, visualized through clear feedback and transparency cues to support trust and acceptance.

6.2 RELEVANCE

The relevance of this work can be divided into multiple levels, concerning users, organizations and the broader transition toward a circular economy. At the user level the prototype demonstrates how DPP-enabled systems can shift from being perceived as compliance-driven reporting tools to becoming practical operational aids, while still following the established DPP framework. By prioritizing usability and reducing cognitive friction, the platform lowers the threshold for engagement and in turn adoption for the technology – the design is therefore optimal for environments where time pressure and task complexity is high.

For organizations involved in remanufacturing the findings highlight that successful DPP implementation is not solely a technical or regulatory challenge, but a design and change management challenge. By aligning with Kotter's (2018) emphasis on creating early wins and visible benefits when introducing new systems, as well as Lewin's (1947) notion of embedding change into everyday practice – this thesis illustrates how front-end design choices such as workflow integration and feedback transparency can directly influence employee acceptance.

At a societal level, the work contributes to ongoing discussions around how the DPP can enable circular economy goals in practice, by allowing user-centered design as a prerequisite for achieving ecological benefits. While regulatory frameworks such as the ESPR define that data must be made available, this thesis illustrates how that available data can be made usable. By focusing on one of the key strategies for reducing environmental impact, remanufacturing, the work supports broader sustainability objectives, including reduced material consumption and lower lifecycle emissions (Zheng et al., 2019).

The relevance of the thesis also lies in its contribution to design knowledge, showcasing how design principles, interaction patterns, and empirically grounded insights can inform future DPP-platforms within RemaNet and similar initiatives. Through this lens, the thesis is positioned as a bridge between policy ambitions, technical DPP infrastructure, and human-centered design implementation.

6.3 REFLECTION

A critical reflection of the project reveals several trade-offs that shaped the outcome. One central decision was to develop three concepts addressing different stakeholder perspectives, while moving further with only one of them into an interactive prototype. This choice reflects a deliberate prioritization of depth over breadth. While this limited the extent to which Concepts B and C could be evaluated – which in turn resulted in the platform not being an all-encompassing solution for the stakeholders – it enabled more rigorous exploration of usability and interaction quality for the primary user group.

The use of an interactive Figma prototype represents another methodological trade-off. The absence of backend programming means that conclusions cannot be drawn about system performance or long-term operational scalability. However, as usability theory suggests, early-stage prototypes are well suited for identifying interaction issues

and evaluating user understanding (Nielsen, 2012). Given the exploratory nature of the research questions, this level of fidelity was appropriate for investigating adoption-related factors rather than technical feasibility. The decision to utilize Google Material's icons and visual language, was highly relevant in terms of learnability and confusion prevention. As Jordan (1998) notes, users rely on familiar patterns to navigate systems efficiently – developing custom icons and visual guides would not only introduce another layer for the user to learn navigate but would also decrease allocation of time to other aspects, such as interaction development. The choice of employing a consistent, already well-known design language would therefore also make the usability evaluation more consistent.

The user testing methodology also introduces limitations. While the tests provided valuable qualitative insights into first-time use, they do not offer statistically generalizable results, as well as results past first-time use (which contributes to valuable user feedback from prolonged use). Nevertheless, this aligns with the thesis' aim to generate design knowledge rather than performance metrics and a finished product through further iteration based on the evaluations. Observing hesitation, confusion, exploration, and verbalized expectations proved particularly effective for uncovering usability gaps related to progress visibility and completion feedback.

Finally, the author's dual role as a designer and researcher necessarily reflexibility. Design decisions are inevitably influenced by the designer's interpretation of theory and data. This risk was mitigated through grounding design choices in established literature, documenting design rationale, building a solid foundation for each concept, and incorporating feedback from supervisors, project partners, and users. Rather than undermining the study, this reflective stance strengthens its contribution by making assumptions and limitations explicit throughout the thesis.

6.4 CONCLUSIONS

This thesis set out to explore how a Digital Product Passport (DPP) platform can be designed to support remanufacturing workflows through a user-centered approach. Rather than treating the DPP as a purely technical or regulatory artifact, the work positioned it as an interactive system whose success depends on usability, trust, and integration into everyday work practices. The results demonstrate that design choices related to information hierarchy and feedback mechanism play a decisive role in whether a DPP platform is perceived as useful or burdensome. Through the development and evaluation of an interactive proof-of-concept prototype, the thesis contributes design knowledge that complements ongoing DPP initiatives. The conclusions below summarize how the objectives were met and how each research question was addressed.

6.4.1 Project Objective and Aims

The overall objective, to explore how a DPP platform can support remanufacturing workflows (through user-centered design), was fulfilled through a combination of theory, empirical input, exploration, and iterative design. The project successfully bridged usability theory and change management perspectives with practical prototyping for an industrial context. The specific aims were addressed as follows:

- Investigating adoption barriers was achieved through exploring the context through benchmarking, interviews, literature review, and user testing – revealing that perceived administrative burden, unclear system state, information overload, and poor workflow alignment are key obstacles.
- Applying user-centered design principles was demonstrated through workflow-based

interface structuring, progressive disclosure of information, layout hierarchy, and mobile-first interaction design (prioritizing core content and essential functions for small screens).
- Exploring data structuring and visualization was done through role-adaptive concepts and a detailed operational prototype, addressing how complex lifecycle data can be made actionable.
- Developing and evaluating a proof-of-concept was accomplished through an interactive prototype of Concept A, which was later validated through user testing with multiple participants from relevant industrial contexts.

These outcomes support the thesis' contribution as design-oriented knowledge and guides for future development within DPP related initiatives.

6.4.2 Research Question 1 (Main)

How should the DPP platform be structured, visualized, adapted, and communicated to support user trust, efficiency, and acceptance?

The results indicate that the most significant adoption barriers are not rooted in resistance to sustainability goals or digitalization itself, but in usability-related challenges. Apart from the introduction of new technology or platforms, stakeholders face difficulties when systems require excessive manual input, present information without context, present irrelevant information, or disrupt established workflows. These findings align with Norman's (2013) argument that poorly designed systems shift cognitive effort onto users, leading to frustrations and avoidance. User testing revealed that even when terminology and data were perceived as credible, uncertainty about task completion and next steps reduced user confidence. This highlights visibility of system status (Nielsen, 1994) and workflow transparency as critical usability factors. The results of the user testing also support CIRPASS findings that adoption depends on minimizing perceived administrative burden and aligning digital tools with existing practices.

6.4.3 Research Question 2

Which principles of user-centered design and change management can be applied to reduce these barriers and enhance adoption?

The thesis demonstrated that several principles are particularly effective in remanufacturing contexts:

- Workflow alignment, supporting user's mental models (Norman, 2013),
- Progressive disclosure, reducing cognitive load (Sweller, 1988),
- Immediate feedback, reinforcing trust and perceived system reliability (Nielsen, 1994),
- Compatibility with existing practices, increasing perceived relative advantage (Rogers, 1983).

From a change management perspective, the prototype supports adoption by making benefits visible early and reducing the learning threshold. Rather than forcing behavioural change the platform, the platform lowers resistance by adapting to existing remanufacturing routines. This aligns with Rogers' (1983) and Kotter's (2018) emphasis on perceived usefulness and early reinforcement.

6.4.4 Research Question 3

How should the DPP system be structured, visualized, adapted, and communicated to support user trust, efficiency, and acceptance?

The findings suggest that a DPP-platform, or system, should be structured around tasks rather than sets of data, be visualized through context-specific views, and communicated via clear system feedback, as well as data transparency indicators (via for example timestamps and signatures). Trust was strengthened when users could

easily see what needed to be done and how their actions contributed to the overall process. The role-adaptive concept approach further indicates that acceptance increases when stakeholders are exposed to responsibility-relevant information only. This is a key factor in interaction design that Preece et al. (2015) refers to as relevance and actionability over completeness. The prototype illustrates how DPP-data can be framed as an operational support, rather than abstract compliance information.

6.5 RECOMMENDATIONS FOR FUTURE WORK

The results of this thesis demonstrate the potential of a user-centered *Digital Product Passport* (DPP) platform to support remanufacturing workflows. Sever factors have however been identified to improve usability and enable future implementation in industrial contexts.

Extended User Testing and Role-Specific Evaluation

Future work should include a broader and more role-specific user studies. While this study did involve usability testing with participants from repair and engineering contexts, additional as well as prolonged testing with specifically remanufacturing operators and managers would provide deeper insight into role-dependent needs and expectations. This view can also be extended to cover all relevant stakeholders such as OEM representatives and end-users (customers), if the need of an all-encompassing platform is still relevant. Longitudinal testing in real work environments would be particularly valuable to assess learnability, efficiency over time, sustained adoption, and other user experience related factors.

Refinement of Feedback and System Status Communication

The user testing that was performed on the interactive prototype indicated a need for clearer feedback regarding task completion regarding workflow progression and system state. Future iterations of the prototype should explore further visual and interaction patterns for communicating progress and "what's next" – such as clearer step indicators, more elaborate progression bar than the current version, toasts of what has been performed as well as what will be next, guides or tutorials, confirmations, and error prevention mechanisms. Refining these elements could reduce initial uncertainty and improve confidence during first-time use, a critical element for adoption in operational settings.

Integration of Live Data and Backend Logic

As mentioned previously, due to time, scope, research questions, and limitations of the prototyping software (Figma), the current prototype only simulates system behaviour without live data connections or backend functionality. To move toward implementation, future work should investigate not only a fully functional backend system for handling, and verifying live data, but also technical integration with existing systems such as ERP and PLM platforms. This would enable evaluation of performance and interoperability in practice, as well as assessment of how real-time data availability affects user trust and decision-making.

Expansion of Concept B and C

Continuing the point in *Extended User Testing and Role-Specific Evaluation*, concepts B and C were only explored at a conceptual level in this thesis. Future research could continue the progression of their development and move them forward into interactive prototypes and evaluate them with relevant stakeholder groups. The concepts could then be united into an all-encompassing platform solution

that link the stakeholders concerning remanufacturing. Separate studies on each concept, however, would allow for comparative analysis of how stakeholder perspectives influence the structuring and communication of DPP data, and how a role-adaptive platform could be designed holistically.

Regulatory Alignment and Standard Evolution

As the DPP is still under development and subject to evolving EU regulations, future work should continuously align design solutions with updated policy frameworks and standards. The recommendation is that the platform remains in development under projects such as RemaNet for the foreseeable future. Collaboration with regulatory bodies and standardization initiatives could help ensure that design concepts remain relevant and compliant as requirements and the technology mature.

Organizational Change and Implementation Strategy

Finally, future studies should explore organizational and change management aspects of DPP implementation in greater depth. Relevant aspects would be training strategies, onboarding flows, leadership strategies, and internal communication practices that support transition from existing systems to DPP-enabled platforms. Investigating these factors alongside front-end and interface design would provide a more comprehensive understanding of how digitals can successfully support circular economy initiatives in industry.

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8 Appendix

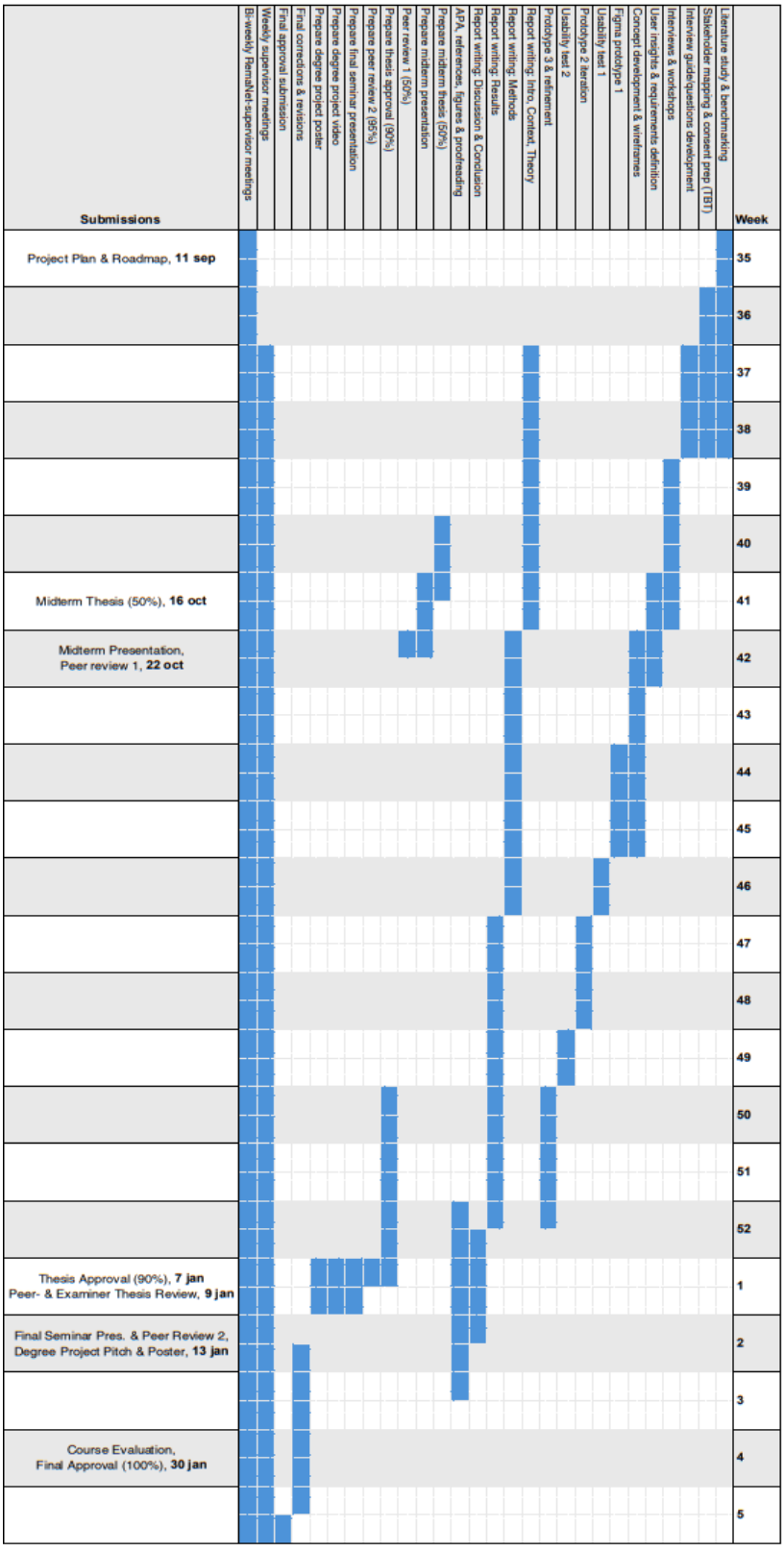
8.1 APPENDIX A: RESULTS OF BENCHMARKING

Benchmark of existing DPP/circular economy-related platforms/frameworks.

Platform / Initiative	Sector / Scope	Key Features	Lessons Learned
Catena-X	Automotive value chain	Federated data exchange, Gaia-X compliance (Gaia-X, 2025), standard APIs	To scale multi-stakeholder platforms *interoperability*[4] and open standards are essential
EURATEX / TRICK	Textile & apparel value chain	Focus on textile traceability, circularity data models	Being sector-specific increases adoption and relevance
Global Battery Alliance	Battery production, reuse, recycling	Defines global battery standard, lifecycle emissions tracking	Granular/component-level data allows for better end-of-life decisions
Siemens Battery Passport	Battery lifecycle management	Real time data exchange, user friendly UI	Seamless integration with existing enterprise systems (Catena-X, etc) improves user adoption
Circularise	Cross-sector transparency & traceability	Blockchain-enabled transparency, material traceability	Trust-building and data sovereignty (blockchain) are critical to stakeholder participation
Arianee	DPP for luxury goods / fashion	Brand-driven DPP, consumer engagement focus, eCommerce	User engagement and good UX can turn compliance into a value-adding experience
BAMB2020	Construction industry & building materials	Building logbooks, material passports, circular construction pilots	Pilots demonstrate that standardized data formats reduce waste and enables reuse
EPREL	Energy-related products (EU-wide)	Centralized product database for EU energy-labelled products	Centralized data access builds transparency and simplifies compliance
CIRPASS / CIRPASS-2	Cross-sector	Blueprint for EU DPP system, data model definitions, governance recommendations	Harmonized governance is needed to make DPPs cross-sector and future proof

[4] The ability for different systems and organizations to seamlessly work together is referred to as interoperability (Lindemulder & Kosinski, n.d.).

8.2 APPENDIX B: GANTT CHART



8.3 APPENDIX C: EXPERT INTERVIEW GUIDE

Introduction

Hello and thank you for taking the time to participate in this interview! My name is Mohammad and I am a master's student in Industrial Design Engineering at Luleå University of Technology. This interview is a part of my thesis, which is conducted within the European project RemaNet. The goal of my thesis is to design a concept and guidelines for a Digital Product Passport (DPP) platform, with focus on making the platform easy to use and valuable for remanufacturers and other stakeholders. During this session I'd like to learn about your current workflows, the challenges you face when accessing product information, and what would make a future platform helpful and easy to adopt in daily work. Your answers will be treated anonymously, specific sensitive information will not be shared in my report. This interview should take around 20 minutes, and there are no right or wrong answers. I am interested in your experience and opinions, and if you don't know the answer to something or don't want to answer that is totally fine. Shall we get started?

Current Practices & Pain Points

1. How does your remanufacturing process work daily?
2. What products do you mostly remanufacture?
3. How do you currently decide whether a motor should be remanufactured, repaired, or scrapped?
4. What kind of product information do you typically receive with incoming products?
5. What data gaps cause delays or errors in your process?
6. How much time is currently spent searching for manuals, specs, or history data?

Data Needs

1. Which of the following data points would be most helpful to have digitally?
    a. Operating hours / cycles
    b. Maintenance history
    c. Disassembly instructions
    d. Hazardous material warnings
    e. Performance specifications
    f. CO_2 savings potential
2. Are there any additional data fields you would want included?

Platform & Access Preferences

1. Would you prefer a standalone platform or an integration with your existing ERP/PLM tools? What tools do you use today?
    a. How would you like to be presented with the disassembly information?
2. Who in your company would use this platform (operators, technicians, engineers, managers)?
3. Which devices are typically available on the shop floor (PC, tablet, hand scanner, smartphone)?
4. How important is offline access (PDF, printable sheets, paper)?

User Experience & Adoption

1. What would make you more likely to use such a platform daily?
2. What would be a deal-breaker (for example: too many clicks, slow loading)?
3. Would you be willing to log remanufacture events into the passport? If so, what is the minimum-effort solution you'd accept (QR scan, one-click logging)?
4. How much time can you spend on onboarding/training for a new system (minutes, hours, days)?

Interoperability & Scaling

1. How critical is it that the platform works for multiple OEMs in the same interface?
2. Would you adopt a DPP system if each OEM required you to use a different portal?

Closing

- Do you have any other ideas or frustrations related to product access?
- Thank you for participating, please let me know if you would like to take part of the results when this project is finished!

8.4 APPENDIX D: USER TESTING PROTOCOL

För varje steg:

1. Vet användaren vad hen ska göra?
    a. Syns nästa logiska steg?
2. Ser användaren rätt knapp?
    a. Är den synlig, tydligt namngiven?
3. Förstår användaren vad kopplingen är?
    a. Matchar den användarens förväntningar?
4. Förstår användaren feedbacken?
    a. Är det tydligt om något lyckades eller misslyckades?

Uppgifter:

1. Vet Skapa ett nytt Project och skanna en Product
2. Hitta nedmonterings instruktionerna till den Product du skannat.
3. Välj en Condition-status för en Component du har monterat från produkten. (t.ex. stator godkänd (Good Condition)).
4. Skicka kullagret till Replacement, resterande komponenter kan du skicka till Cleaning.
5. Ersätt kullagret med ett kullager från ditt lager (skicka sedan den till testing), resterande komponenter kan du tvätta.
6. Hitta instruktioner för att montera ihop produkten igen.
7. Gör ett slut test på den färdigmonterade produkten och avsluta sedan projektet.

Insamling av data

Medans uppgifterna görs:

- Var deltagaren tvekar,
- Felklick,
- Uttalanden som: "Jag trodde det skulle vara här",
- Misstolkningar av ikoner/termer,
- När användaren letar runt på skärmen.

Fråga efter varje uppgift (1-5):

- Hur säker kände du dig i denna uppgift?
- Vad förväntade du dig skulle hända härnäst?

Avslutande frågor

Information & visualisering

- Var det för mycket eller för lite information på någon sida?
- Var något otydligt eller visuellt rörigt?
- Kändes terminologin korrekt för din industri/bransch?

Navigation

- Kände du alltid att du visste vart du skulle klicka?
- Var något oväntat eller ologiskt?

Tillit & Transparency

- Kändes systemet pålitligt?
- Fanns det något som minskade din tillit?

Nytta

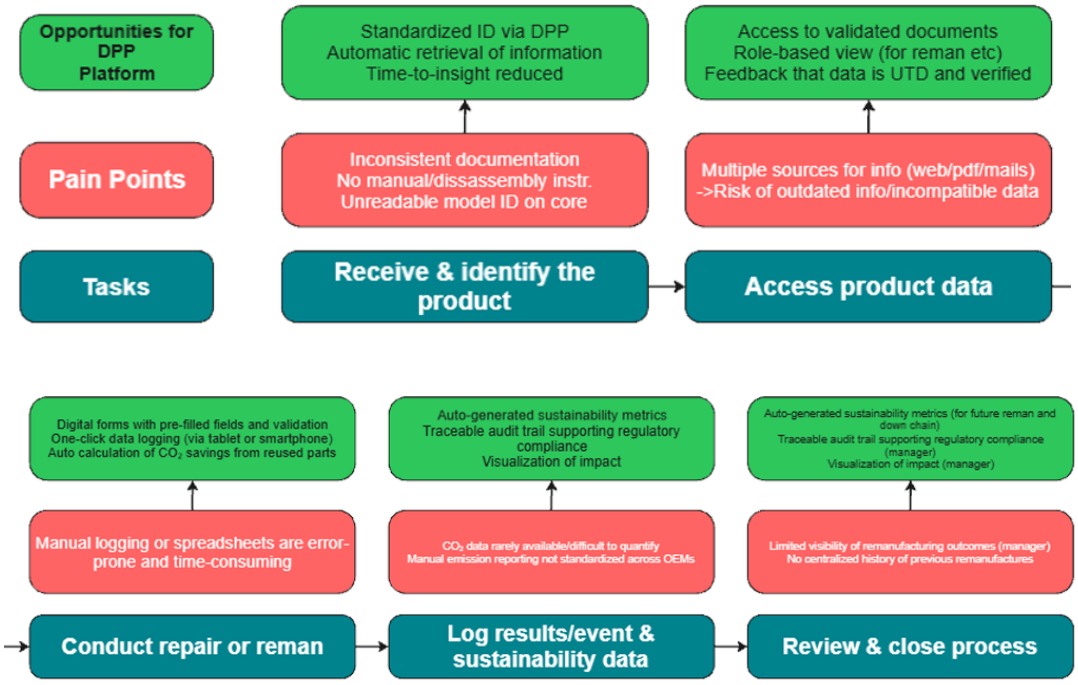
- Skulle detta verktyg underlätta ditt arbete?
- Vad skulle du ändra först?

Adoption

- Hur troligt är det att du skulle använda systemet regelbundet? (mellan 1–5)
- Vad skulle hindra dig från att börja använda det?

Avslut

8.5 APPENDIX E: USER JOURNEY



8.6 APPENDIX F: TABLE OF COMPONENTS

Concept A

Page 1: Core Collection

Category	Component Examples	Function
Input Control Elements	QR-code input, manual serial number field, dropdown (product type), checkbox (core condition: intact/ damaged).	Enables fast registration of incoming cores via scan/manual entry.
Navigational Elements	Location breadcrumbs (dashboard -> collection), "Next Step" button, search bar for existing records.	Guides user through workflow and ensures traceability.
Informational Elements	Confirmation toast ("core logged"), product photo preview, timestamp indicator.	Provides immediate feedback and verification.
Container Elements	Core intake form panel, collapsible section for customer details.	Organizes data into clear/minimal input zones.
Interactive Elements	Upload images of damage parts, clickable checklist for component condition review	Reduces typing effort and ensures visual evidence of condition

Page 2: Inspection and Sorting

Category	Component Examples	Function
Input Control Elements	Dropdown for inspection outcome (Reman / Repair / Scrap), text box for notes, checkbox (Send to disassembly).	Records decisions and reasoning in structured way.
Navigational Elements	Tabs (Visual inspection, Functional test), "Previous / Next" buttons.	Allows switching between inspection modes efficiently.
Informational	Alert banner if	Prevents data gaps and

Elements	incomplete data, tooltips explaining defect categories.	misclassification.
Container Elements	Visual defect log area, image carousel for inspection photos.	Consolidates visual evidence and data for decision review.
Interactive Elements	Clickable rating icons (for instance, damage severity), upload defect photos of product.	Enhances precision of defect reporting through visual input.

Page 3: Strip Core (Disassembly)

Category	Component Examples	Function
Input Control Elements	Step-by-step disassembly checklist, toggle (Part reusable/remanufacturable?), dropdown (Component ID).	Ensures standardized logging of part removal.
Navigational Elements	Process progress bar, index of components.	Supports tracking and navigation through multiple parts.
Informational Elements	Real-time safety warnings (hazardous material), instructional icons.	Reinforces safe and compliant operation.
Container Elements	Accordion layout for component groups (electrical / mechanical).	Prevents interface clutter during large assemblies.
Interactive Elements	Embedded disassembly diagram/instructions with clickable parts, tool overlay for tool references (and torque specs).	Provides interactive visual aid and reduces errors.

Page 4: Cleaning and Refurbishing

Category	Component Examples	Function
Input Control Elements	Checkbox list of cleaning actions completed, dropdown for process used (chemical etc.).	Tracks standard cleaning protocols.
Navigational Elements	Breadcrumbs, "Next step: Testing" button.	Guides workflow continuity.
Informational Elements	Notification ("All components cleaned successfully"), emissions tracker based on components saved.	Gives progress and motivational sustainability feedback.
Container Elements	Panel grouping reusable vs. non-reusable	Provides overview of resource recovery efficiency.

	components.	
Interactive Elements	Drag-and-drop sorting of cleaned parts into status areas (cleaned/send to remanufacture/scrap).	Adds a tactile, intuitive layer to digital tracking.

Page 5: Component Testing

Category	Component Examples	Function
Input Control Elements	Numeric entry for test values, pass/fail toggle, dropdown for testing method.	Records test outcomes consistently.
Navigational Elements	Tabs (Electrical / Mechanical tests), back and forward buttons.	Ensures easy access to related test sections.
Informational Elements	Real-time result comparison to specification ranges, alert if value out of range.	Prevents manual errors and ensures quality control.
Container Elements	Test result table with expandable rows for details.	Structures technical data clearly for review.
Interactive Elements	Interactive graph plotting performance trends.	Visualizes results for quick interpretation.

Page 6: Reassembly

Category	Component Examples	Function
Input Control Elements	Checkbox list for assembly steps, dropdown for part source (new / remanufactured), toggle (Torque verified).	Ensures assembly data and component traceability are properly documented.
Navigational Elements	Breadcrumb trail (Testing -> Reassembly), "View exploded diagram" button, "Next: Final Test" link.	Maintains workflow continuity and supports navigation between detailed references.
Informational Elements	Assembly progress indicator, confirmation alerts for completed steps, warnings for missing components.	Provides transparency and reduces the risk of assembly errors.

Container Elements	Split view window: left for assembly instructions, right for live data input.	Allows parallel reference and logging without disrupting workflow.
Interactive Elements	3D model or schematic viewer with clickable components, real-time validation when each step is logged.	Supports efficient assembly guidance and reinforces procedural accuracy.

Page 6: Final Testing & Quality Assurance

Category	Component Examples	Function
Input Control Elements	Numeric entry fields for test readings, dropdown for test type, pass/fail buttons, signature box for quality approval.	Captures verification data in a standardized and auditable format.
Navigational Elements	Tabs for test categories (Performance, Safety, Compliance), navigation to Generate Report.	Enables structured access to test procedures and results.
Informational Elements	Live status panel ("Test in progress / passed / failed"), toast notifications for saved data, summary score.	Ensures users receive immediate feedback on process outcomes.
Container Elements	Dashboard summary with expandable sections (for example, mechanical test results, CO_2 data, operator notes).	Organizes detailed testing data for review and certification.
Interactive Elements	Interactive chart comparing pre- and post-reman performance, export function to PDF / DPP record.	Facilitates visual verification and efficient documentation.

Concept B

Page 1. Overall Status Dashboard.

Category	Component Examples	Function
Header & Navigation	Logo & system title, Global navigation bar, Profile/account	Provides access to system-wide sections; ensures

		consistent navigation.
Global KPIs	Total products in remanufacturing today, Average turnaround time, Current backlog (# of items), On-time delivery %	Shows the operational "pulse" of remanufacturing immediately.
Status Overview Tiles	"In Progress" count, "Awaiting Parts" count, "Testing Stage" count, "Ready for Shipment" count	Allows quick assessment of job distribution across stages.
Notifications Panel	Alerts for bottlenecks, Technician support requests, Delayed jobs, Missing data warnings	Surfaces urgent issues without navigating into detailed pages.
High-Level Sustainability Snapshot	CO_2 saved today/this week, Material recovery %, Reused parts vs. new parts ratio	Gives the manager immediate sustainability feedback without needing to open the CO_2/impact page.
Recent Activity Feed	Newly logged DPP events, completed reman jobs, Flagged anomalies	Provides traceability and understanding of what happened recently.
Quick Filters	Filter dashboard by product family, site, technician, date	Enables fast contextual adjustments without switching pages.

Page 2. CO_2 & Sustainability Metrics

Category	Component Examples	Function
Header & Navigation	Page title ("Sustainability & CO_2 Metrics"), Breadcrumb ("Dashboard > Sustainability"), Profile/settings	Provides consistent structure and quick path back to dashboard.
High-Level Environmental KPIs	Total CO_2 saved (week/month/year), Material recovery %, Reuse vs. replace ratio, Avg. ROI of remanufacturing	Offers a strategic, aggregated overview of environmental performance.
Trend Charts & Graphs	CO_2 savings over time, Recovery rate trends, Waste reduction curve, Energy efficiency improvements	Supports time-based analysis to identify long-term improvements or declines.
Per-Product-Family Breakdown	Electric motors, Drives, PLCs, Sensors	Shows sustainability impact by category, enabling targeted optimization.
Job-Level Impact Cards	Individual remanufacturing job CO_2 impact, Material reuse amount, Parts replaced vs. repaired, Energy savings	Allows drilling down into specific remanufacturing activities.
DPP-Based Environmental	Material composition, Hazardous materials indicators, End-of-life	Displays environmental information retrieved

Data Panel	recommendations	directly from DPP entries.
Filters & Segmentation Tools	Time window (week/month/year), Product type, OEM, Facility/site, Technician/team	Enables tailored sustainability views for strategic reporting.
Export & Reporting Tools	Export chart as PDF, Export raw CO_2 data as CSV, "Add to Sustainability Report" button	Supports external reporting requirements and stakeholder communication.
Comparison Mode	Compare two time periods, compare two product families, Compare two OEMs	Enables performance benchmarking internally or externally.

Page 3. DPP Data Quality & Completeness.

Category	Component Examples	Function
Header & Navigation	Page title ("DPP Data Quality & Completeness"), Breadcrumb ("Dashboard > Data Quality"), Profile/settings	Ensure consistent layout and easy navigation back to other pages.
Overall Data Quality Score	Circular progress indicator (for example: "82% complete"), Color-coded quality level (green/yellow/red)	Provides a high-level summary of how complete and reliable imported DPP data currently is.
Data Completeness by Category	Material composition, Hazardous substances, Disassembly instructions, Repair history, Environmental footprint data	Shows which types of data are missing or inconsistent across products.
OEM Data Reliability Matrix	Table showing OEM name, % completeness, last update, known data gaps	Helps identify which OEMs supply well-structured DPP data and which require follow-up.
Flagged Records	List of items with critical data gaps, Missing disassembly steps, Inconsistent part IDs, Outdated environmental declarations	Highlights DPP entries that require review before they cause workflow delays.
Per-Product DPP Audit Cards	"Electric Motor-DPP Score: 75%", "Drive Module-DPP Score: 92%"	Allows drilling down into the data integrity of specific product types.
Data Validation Tools	"Refresh data from OEM" button, Conflict resolution prompts (for example: mismatched component IDs), Data version comparison	Gives managers control over maintaining up-to-date and consistent DPP entries.
Filtering & Segmentation	Filter by OEM, Filter by product category, Filter by severity (critical/major/minor)	Helps users focus on the most urgent or relevant data issues.
Export Tools	Export missing data list (CSV), Export OEM data audit report (PDF)	Supports communication with suppliers, regulatory

		reporting, and internal quality teams.

Page 4. OEM & Supplier Performance.

Category	Component Examples	Function
Header & Navigation	Page title ("OEM & Supplier Performance"), Breadcrumb ("Dashboard > OEM Performance"), Profile/settings	Provides clear orientation and allows return to the general dashboard.
OEM Performance Overview	Ranking list of OEMs, Badges showing "Gold / Silver / Bronze" performance, KPIs summarizing on-time data delivery	It gives managers a quick overview of which OEMs provide the most reliable data and cooperation.
OEM Performance Scorecards	DPP Data Accuracy %, Data Completeness %, Average Response Time to Data Requests, CO_2 Data Availability, Disassembly Data Coverage	Shows multidimensional metrics for each OEM to support procurement, communication, and escalation.
Trend Graphs	"OEM Data Quality Over Time" line graph, "Response Time Trend (last 12 months)", "Completeness Trend by Category"	Visualizes if OEM collaboration is improving, declining, or stable over time.
Compliance & Regulation Status	CE marking confirmation, ESPR compliance flags, Missing mandatory DPP fields list	Help assess which OEMs pose compliance risks or require follow-up before audits.
Supplier Communication Hub	Pre-written request templates ("Request missing disassembly instructions", "Request updated environmental data"), Message history with each OEM	Enables managers to communicate data gaps directly and track correspondence.
Supplier Risk Indicators	Risk level tags (Low / Med / High), Reasons (for example: "Unresponsive", "CO_2 data missing")	Highlights suppliers that may jeopardize remanufacturing quality, or regulatory compliance.
Filter & Compare Tools	Filter by OEM, Filter by risk level, Compare two OEMs side-by-side	Supports strategic evaluation and procurement decisions.
Export Tools	Export supplier performance report (PDF), Export compliance audit summary (CSV)	Enables reporting to management, sharing with procurement teams, or preparing for regulatory checks.

Page 5. CO_2 & Sustainability Insights.

Category	Component Examples	Function
Header &	Page title ("CO_2 & Sustainability	Clarifies location in the

Navigation	Insights"), Breadcrumb ("Dashboard-Sustainability"), Profile/settings	platform and supports navigation.
Top-Level Sustainability KPIs	Total CO_2 saved (YTD), CO_2 saved per product category, Remanufacturing rate (%), Waste diverted from landfill (kg), Energy savings equivalent	Provides a high-level sustainability summary for managers and reporting.
CO_2 Savings Visualization	Line graph: "CO_2 Savings Over Time", Bar chart: "CO_2 Savings by Product Type", Pie chart: "Emission Sources Distribution"	Enables quick pattern recognition and supports ESG reporting.
Reman vs New Production Comparison	"Environmental Benefit per Unit" score, CO_2 avoided by reman vs new manufacturing, Energy consumption reduction, Water usage reduction	Highlights the environmental impact of remanufacturing to support decision-making and communication.
Lifecycle Impact Breakdown	Cradle-to-gate CO_2 footprint for components, Indicators such as material extraction, manufacturing, transport, remanufacturing processing, end-of-life	Help managers identify components with the highest environmental impact.
Sustainability Alerts	Missing CO_2 data from OEMs, Incomplete sustainability fields, Upcoming reporting deadlines, Outliers in CO_2 emission values	Supports proactive management for compliance and internal reporting.
ESG Documentation & Export Tools	"Generate ESG Snapshot (PDF)", "Export CO_2 dataset (CSV)", "Download component-level LCAs (if available)"	Provides data needed for audits, stakeholder communication, and sustainability reports.
Filter Controls	Filter by date (week/month/year), Filter by product type, Filter by remanufacturing site, Filter by OEM	Supports focused analysis and comparison.
Insight Cards	"Top 5 products by CO_2 savings", "Lowest-performing products (highest footprint)", "Biggest improvement over last 6 months"	Surfaces the most actionable sustainability insights.
Scenario Simulation (Optional)	Dropdown: "If reman rate increased by X%…", Visualization showing potential CO_2 impact, "Compare scenarios"	Helps managers model how operational changes affect sustainability outcomes.

Concept C

Page 1. Home / My products.

Category	Component Examples	Function
Informational Elements	Product list cards, product thumbnail, product name/ID, current status label	Provides an overview of all customer-owned products registered in the platform.
Navigational Elements	"View Progress" button, product card click area	Allows the user to navigate to detailed progress views for each product.
Input Control Elements	"Send Product" primary action button	Enables the user to initiate a new remanufacturing request.
Container Elements	Scrollable product list, grouped by status	Organizes multiple products in a clear, manageable layout.
Interactive Elements	Status indicators (icons/colors), expandable product cards	Reinforces transparency and gives immediate feedback on product state.

Page 2. Send for Remanufacturing (Submission)

Category	Component Examples	Function
Input Control Elements	Product ID input (scan/UPI/manual), text field for fault description, dropdown for shipping method	Captures essential information required to initiate a remanufacturing request.
Navigational Elements	Step indicator (1–3), "Submit" button, "Back" button	Guides the user through a short, structured submission flow.
Informational Elements	Helper text, confirmation message ("Submission successful")	Reduces uncertainty and reassures the user that the request was completed correctly.
Container Elements	Form sections (Product info/Shipping/Confirmation)	Groups related inputs to simplify interaction.
Interactive Elements	QR scan trigger, selectable shipping options	Minimizes effort and lowers the threshold for initiating remanufacturing.

Page 3. Product Progress

Category	Component Examples	Function
Informational Elements	Step-based progress tracker (Received-Return-Shipping), timestamps, short explanatory text	Communicates remanufacturing status in a non-technical, customer-friendly way.
Navigational Elements	"Back to My Products" button	Allows the user to easily return to the overview.
Visual Indicators	Completed/in-progress/upcoming icons	Provides instant understanding of current state and progress.
Container Elements	Vertical progress timeline, expandable step cards	Structures process information while avoiding information overload.

Interactive Elements	Expand/collapse step details	Lets users choose how much detail they want to see.

Page 4. Sustainability Summary

Category	Component Examples	Function
Informational Elements	CO_2 savings metric, material recovery percentage, component reuse summary	Makes sustainability impact tangible and understandable for the customer.
Visual Elements	Charts, comparison bars (remanufactured vs new), circularity score badge	Reinforces impact through visual storytelling rather than raw data.
Navigational Elements	"Download Certificate" button	Enables export of sustainability data for reporting or documentation.
Container Elements	Summary cards, sustainability highlights panel	Presents key outcomes in a concise and structured format.
Interactive Elements	PDF download trigger	Supports CSRD/ESPR-aligned information sharing.

Page 5. Service History & Documentation

Category	Component Examples	Function
Informational Elements	Timeline of past remanufacturing events, dates, outcomes, CO_2 savings per cycle	Provides a long-term, traceable service history aligned with DPP principles.
Navigational Elements	Filter or sort options (by date, product)	Helps users navigate historical records efficiently.
Document Elements	Warranty information, usage recommendations, authorized DPP fields	Builds trust and supports informed product use.
Container Elements	Expandable history entries, document sections	Keeps long-term data readable and accessible.
Interactive Elements	Document download links, expandable record views	Encourages continued engagement with the platform beyond a single service event.